



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

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Bioinspired and Soft Robotics

Soft Robotics Technologies

Electro-Active Polymers

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History

The field of EAPs emerged back in 1880, when Wilhelm Röntgen designed an experiment in which he tested **the effect of an electrical current on the mechanical properties** (change in length) of a rubber band.

It wasn't until 1925 that the first piezoelectric polymer was discovered (Electret). Electret was formed by combining carnauba wax, rosin and beeswax, and then cooling the solution while it is subject to an applied DC electrical bias.

The mixture would then solidify into a polymeric material that exhibited a piezoelectric effect.

In 1949, Katchalsky et al. demonstrated that when collagen filaments are dipped in acid or alkali solutions they would respond with a change in volume. The collagen filaments were found to expand in an acidic solution and contract in an alkali solution. Although other stimuli (such as pH) have been investigated, due to its ease and practicality most research has been devoted to developing polymers that respond to electrical stimuli in order to mimic biological systems.

In 1969, Kawai demonstrated that polyvinylidene fluoride (PVDF) exhibits a large piezoelectric effect. This sparked research interest in developing other polymers systems that would show a similar effect.

In 1977, the **first electrically conducting polymers** were discovered by Hideki Shirakawa et al.

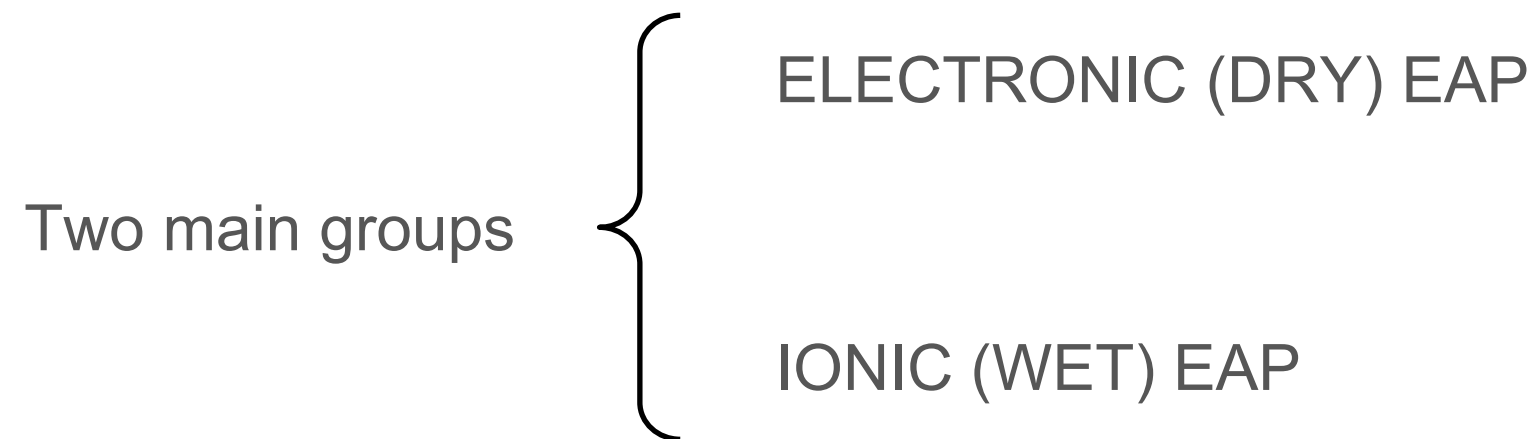
In the early 1990s, **ionic polymer-metal composites were developed and shown to exhibit electroactive properties far superior to previous EAPs**. The major advantage of IPMCs was that they were able to show activation (deformation) at voltages as low as 1 or 2 volts.

DARPA-funded research in the 1990s at SRI International and led by Ron Pelrine developed an electroactive polymer using silicone and acrylic polymers; the technology was spun off into the company Artificial Muscle in 2003, with industrial production beginning in 2008.



Definition

- **Electrically activated** polymeric material
- Include a large number of **different technologies**, in their functional principles as well as in their level of development
- The most interesting characteristic is their capability to **emulate biological muscles** being very similar in elasticity, hardness, stiffness and deformations



Definition

Electronic (dry) EAP

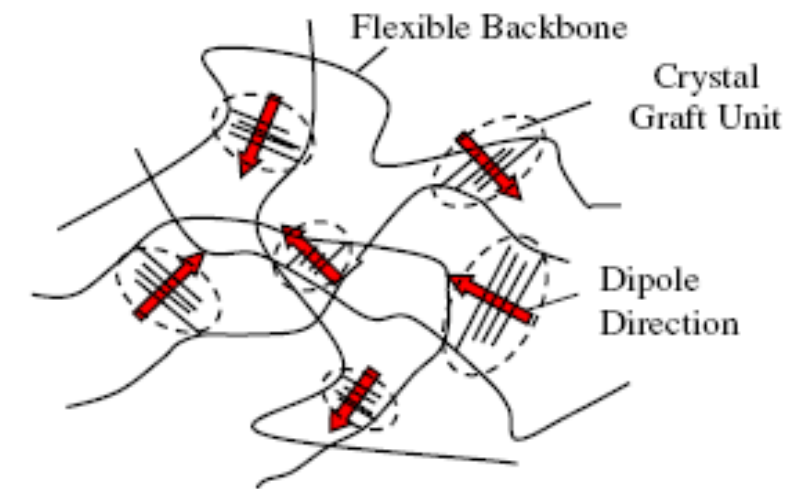
- Electrostrictive Graft Elastomers
- Ferroelectric Polymers
- Liquid Crystal Elastomer (LCE) Materials
- Dielectric EAP

Advantage	Disadvantage
Can operate in room conditions for long periods of time	Requires high voltages (~150 MV/m)
Rapid response time (mSec levels)	Requires compromise between strain and stress
Can hold strains under DC activation	Glass transition temperature is inadequate for low temperature actuation tasks
Induces relatively large actuation forces	



Electrostrictive Graft Elastomers

The flexible backbone chain and the crystal graft unit consist of polarized monomers, which contain atoms with electric partial charges, generating dipole moments. Upon application of an electric field to the polymer, force is applied to each partial charge, forming a force moment.



The electric field causes the deformation due to two mechanisms:

1. Crystals rotation
2. Chain reorientation

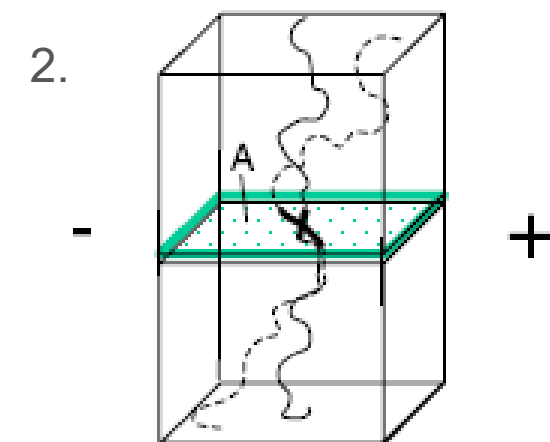
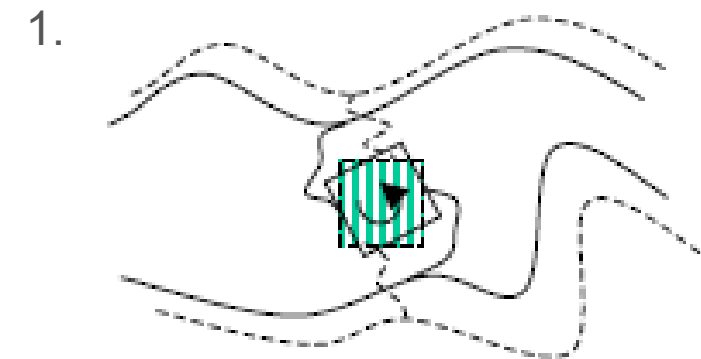
CHARACTERISTICS

- High stiffness ($E=550$ MPa)
- Strain above 5 %
- Relatively high force (22 MPa)
- High voltage required (150 MV/m)
- Energy density: 247 kJ/m^3
- Dielectric constant: 11

MATERIALS

Polymers: Graft elastomer powder in N,N-dimethylformamide

Electrodes: gold



Ferroelectric Polymers

Ferroelectric materials, like ferromagnets in an amorphous matrix, have dipoles that can be aligned producing dimensional changes in response to an applied electric field (electrostrictive effect)

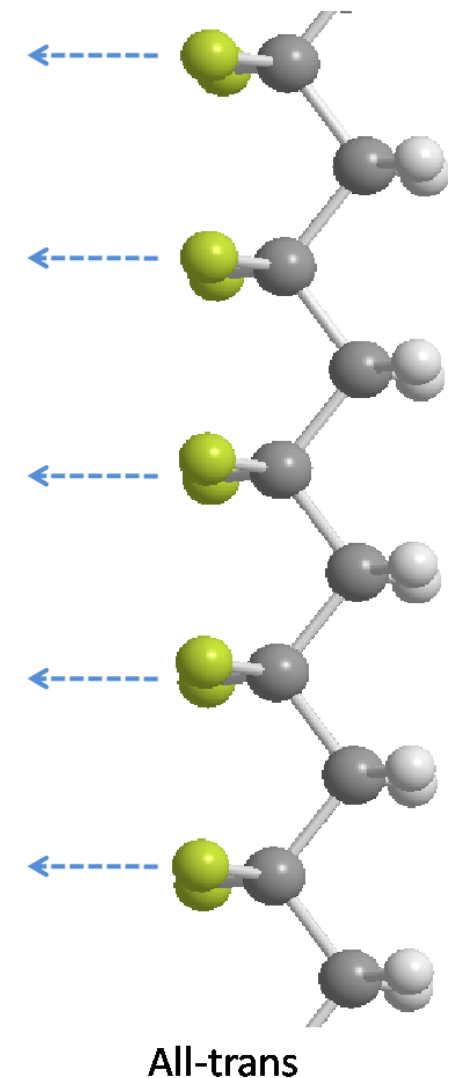
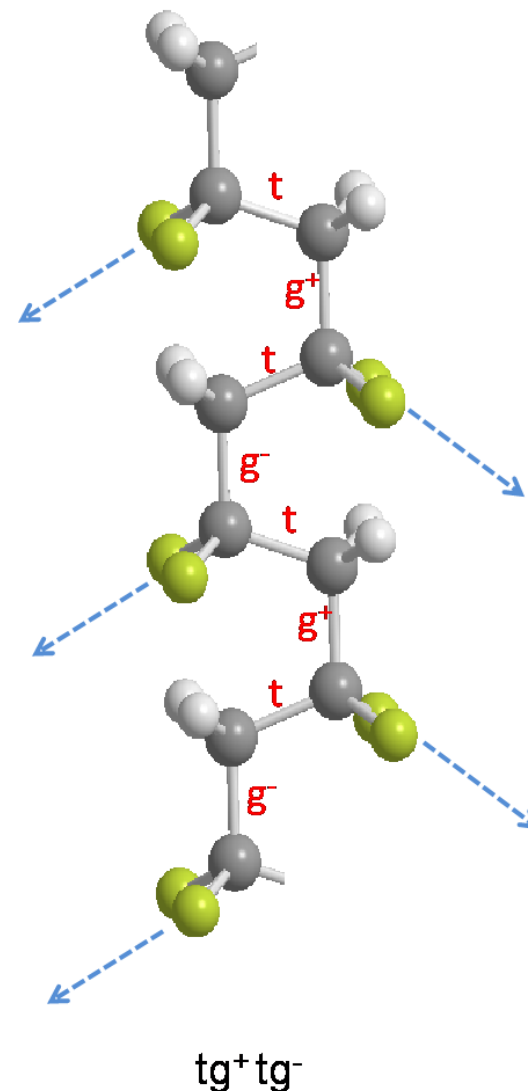
CHARACTERISTICS

- Strain: 1-4,4% (< 7 %)
- Stress: 45 MPa
- Required electric field: 150 MV/m
- Easily found in film
- Low currents
- E=400 MPa
- High hysteresis
- Limited life due to fatigue
- Limited temperature operational range

MATERIALS

Polymers: Poly (vinylidene difluoride) (PVDF) and copolymers P(VDF-TrFE)

Electrodes: gold



Liquid Crystal Elastomer (LCE) Materials

Liquid crystals are known to change phase and orientation under the influence of an electrostatic force (or thermal input).

By incorporating mesogens into a compliant polymer backbone or as side chains, field-induced changes in phase can be used to produce actuation.

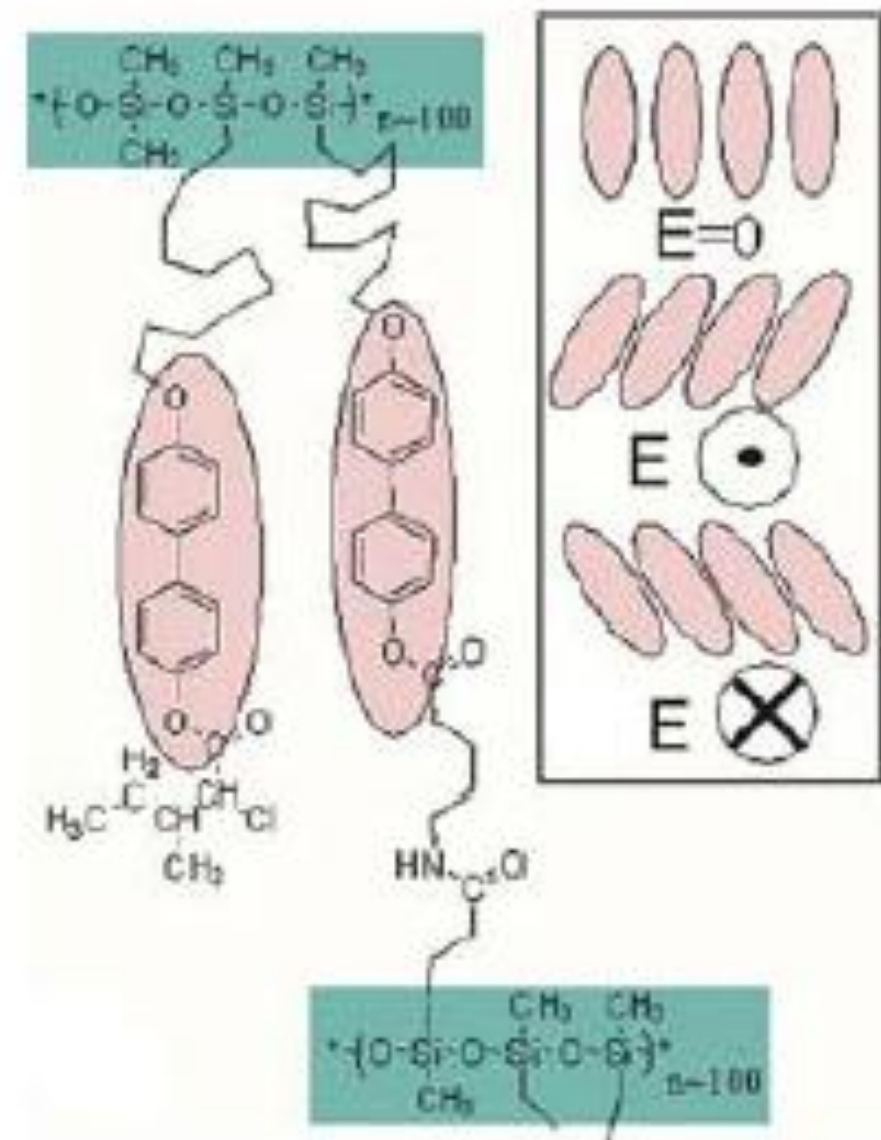
CHARACTERISTICS

- Strain: 5-35 % (30 % siloxane, 10 % acrylic)
- Stress: 100 kPa
- Response time: < 50 ms
- Very slow reverse process requiring cooling
- Required electric field: 1,5 MV/m
- Low stiffness
- Limited tensile resistance

MATERIALS

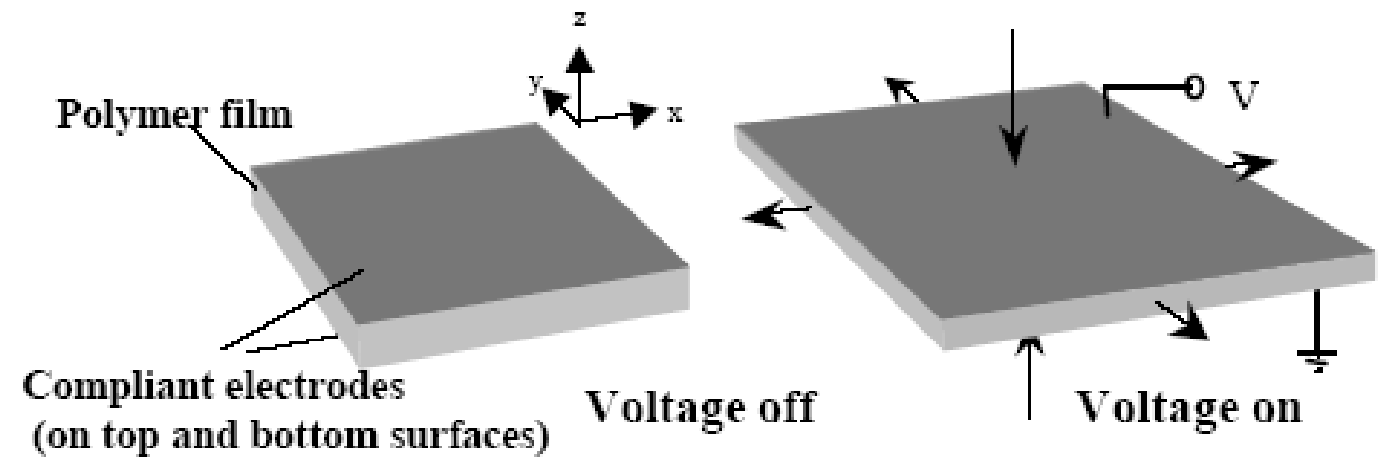
Backbone: Siloxane or acrylic

Cross-linker: Mesogens

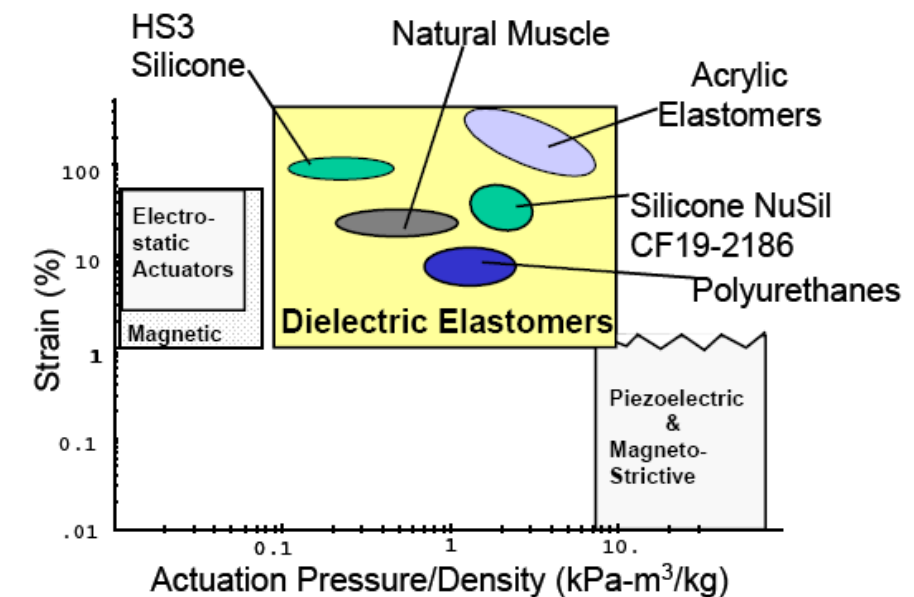


Dielectric EAP

A polymer is sandwiched between two compliant electrodes, which are in intimate contact with the polymer film



Exposed to an electric field, electrodes tend to move closer (for the Maxwell stress effect) and consequently the film undergoes a stretching along the other two dimensions (perpendicular to the field)



CHARACTERISTICS

- Easy to construct and handle
- Many possible configurations
- Large deformations
- Constant volume
- High electric field required

MATERIALS

Polymers: Acrylic
Silicones
Polyurethanes

Electrods: Graphite spray
Carbon grease
Graphite powder
Thickened electrolyte solution
Photopatternable materials



Dielectric EAP

Maxwell stress effect – theoretical model

$$u = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon_r \epsilon_0 \mathbf{E}^2$$

$$U = uAz = \frac{1}{2} CV^2$$

CONSTANT
VOLUME

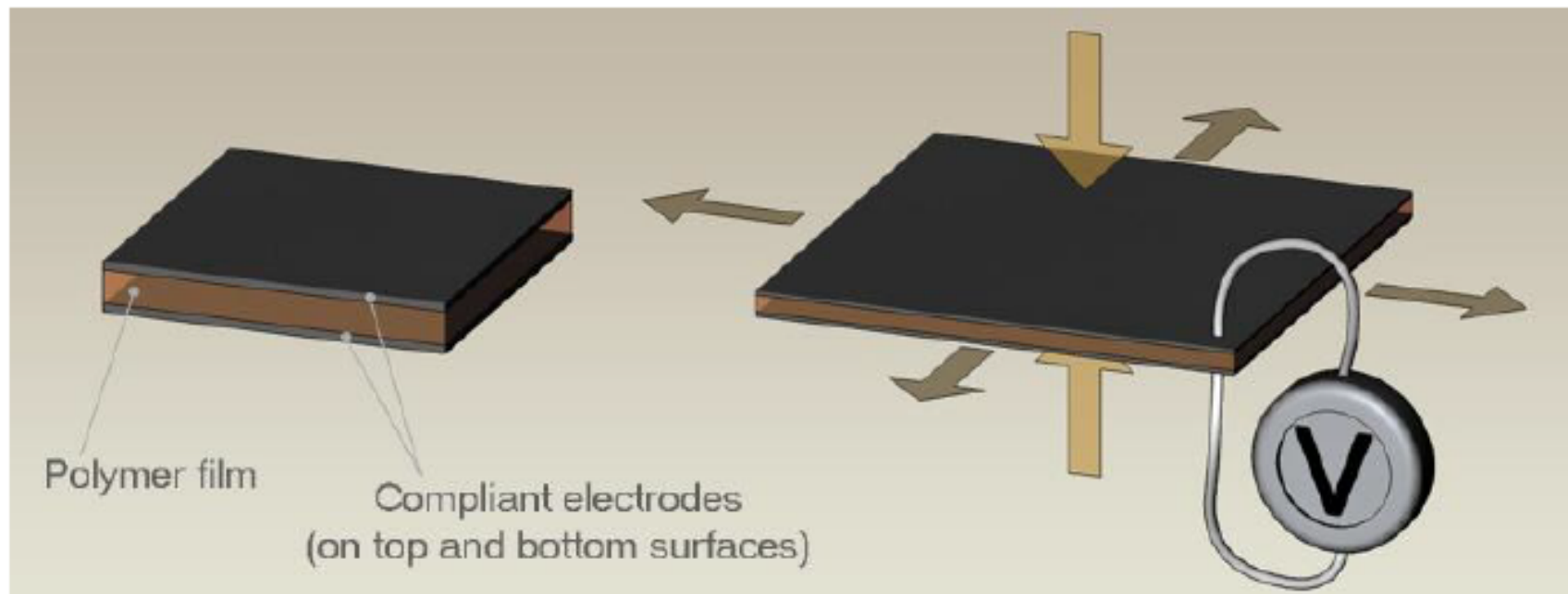
$$\frac{dA}{A} = -\frac{dz}{z}$$

$$dU = -\frac{\epsilon_r \epsilon_0 V^2}{2} 2 \frac{A}{z^2} dz$$

$$F = -\frac{dU}{dz} = \frac{\epsilon_r \epsilon_0 V^2}{z^2} A$$

Stress

$$S = \frac{F}{A} = \frac{\epsilon_r \epsilon_0 V^2}{z^2}$$



Dielectric EAP

Maxwell stress effect – theoretical model

$$u = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon_r \epsilon_0 \mathbf{E}^2$$

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Stress

$$S = \frac{F}{A} = \frac{\epsilon_r \epsilon_0 V^2}{z^2}$$

Polymer film

- Low stiffness
 - High dielectric constant
 - High breakdown strength
- (Acrylic, Silicone, Polyurethane)

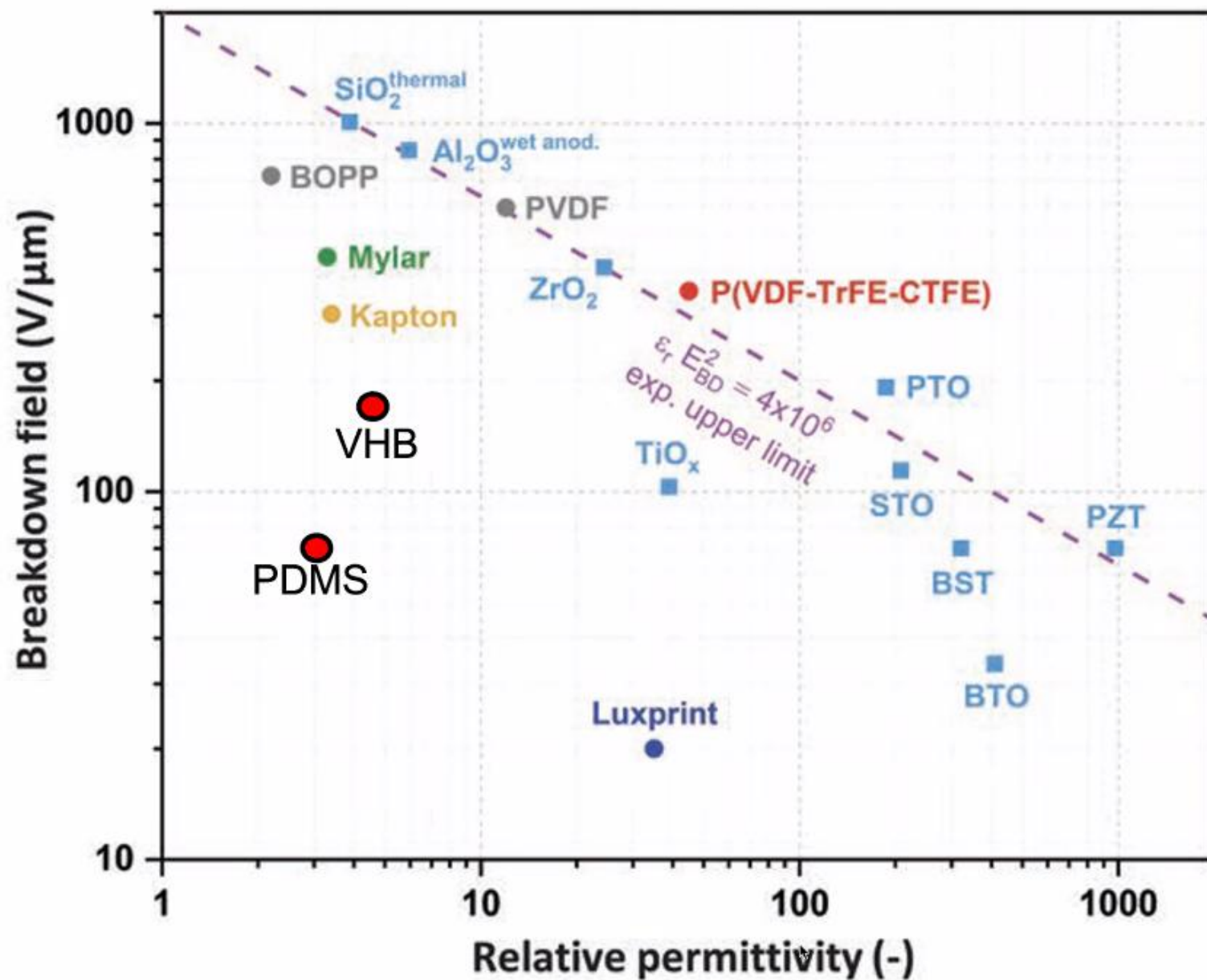
Electrodes

- High compliance
 - High electrical conductivity
 - Negligible contribution to stiffness
 - Low thickness
- (Graphite spray, Carbon grease, Graphite powder, Thickened electrolyte solution, photopatternable materials)



Dielectric EAP

Maxwell stress effect – theoretical model



There is an empirical upper limit to $\epsilon_r E_{BD}^2$

Hinchet, R., Shea, H., High Force Density Textile Electrostatic Clutch. *Adv. Mater. Technol.* 2019, 5, 1900895.



Dielectric EAP

Maxwell stress effect – theoretical model

$$u = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon_r \epsilon_0 \mathbf{E}^2$$

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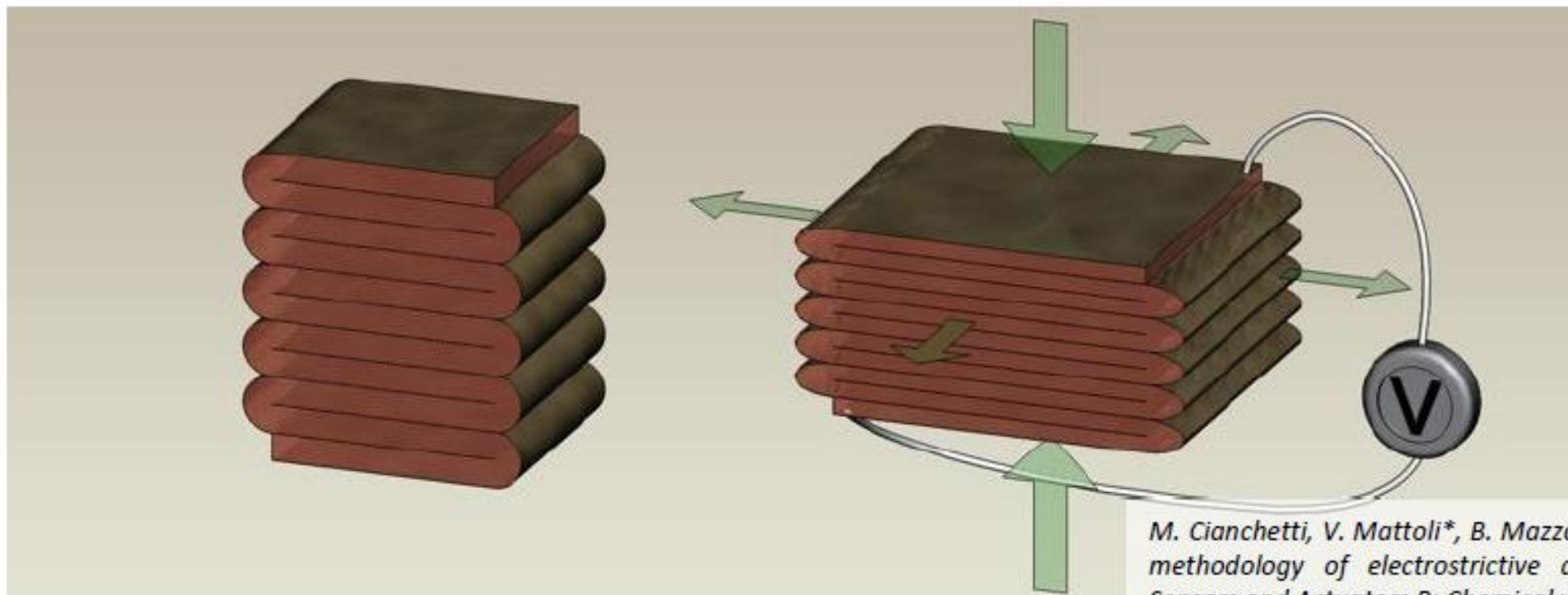
CONSTANT
VOLUME

$$\frac{dA}{A} = -\frac{dz}{z}$$

$$dU = -\frac{\epsilon_r \epsilon_0 V^2}{2} 2 \frac{A}{z^2} dz$$

$$F = -\frac{dU}{dz} = \frac{\epsilon_r \epsilon_0 V^2}{z^2} A$$

$$dU = \frac{\epsilon_r \epsilon_0 V^2}{2} d\left(\frac{A}{z}\right) = \frac{\epsilon_r \epsilon_0 V^2}{2} \left(\frac{1}{z} dA - \frac{A}{z^2} dz\right)$$



Stacked
Actuator
structure

**Folded
Geometry**

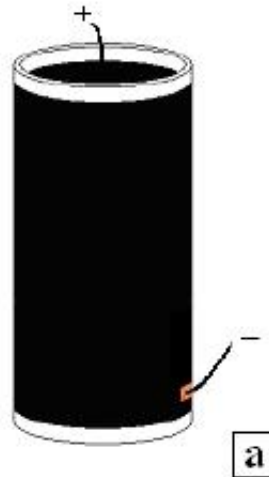
M. Cianchetti, V. Mattoli, B. Mazzolai, C. Laschi, P. Dario, A new design methodology of electrostrictive actuators for bio-inspired robotics, Sensors and Actuators B: Chemical vol. 142 (1), pp. 288-297, 2009*



Dielectric EAP

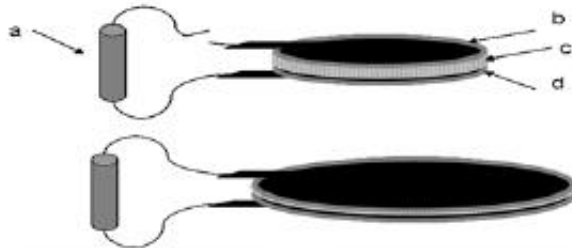
Applications – Different geometries

TUBE (linear actuation)



[VIDEO linear stacked actuator](#)

Circular Plate (radial actuation)

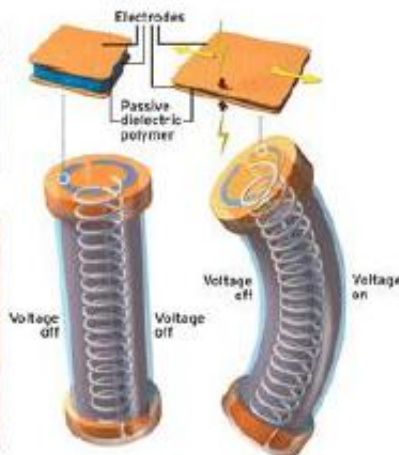
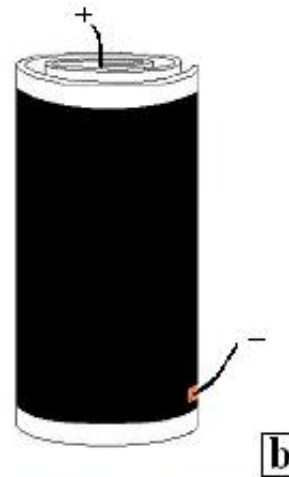


Voltage Off



Voltage On

Spring- ROLLED (bending actuation)

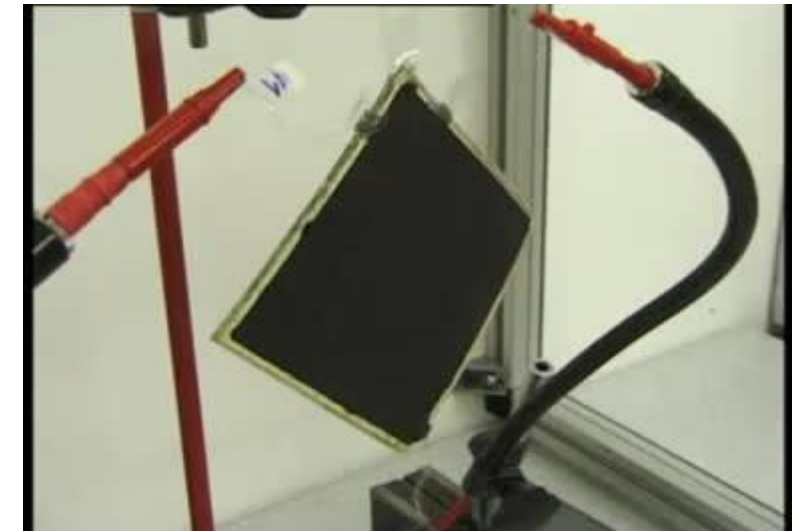
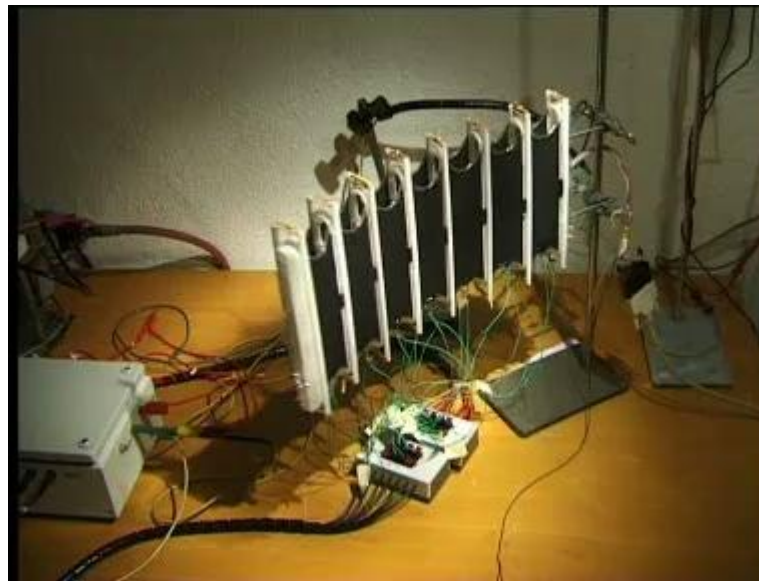
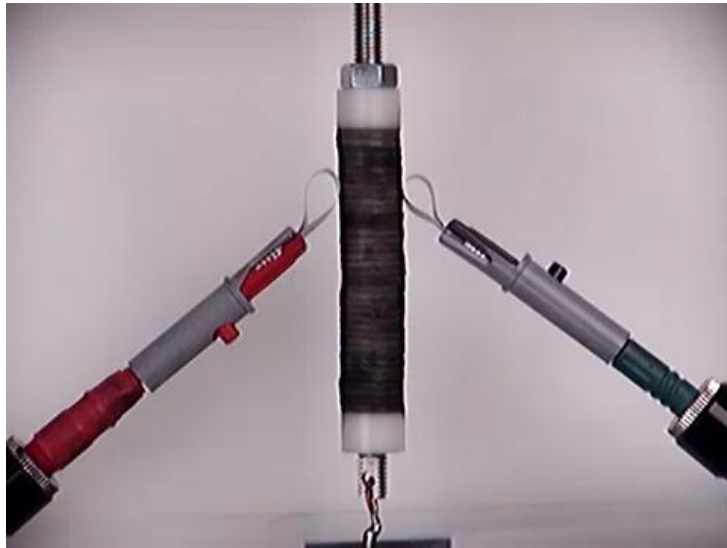


OTHERS (e.g stack, diaphragm)



Dielectric EAP

Applications – Different geometries



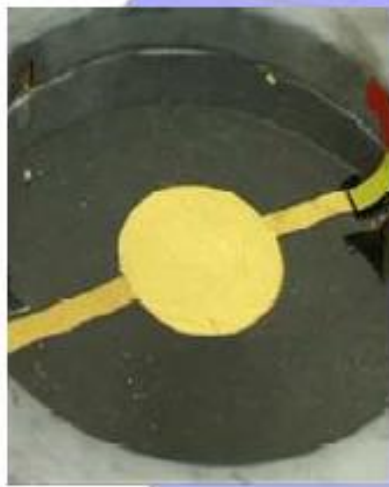
Dielectric EAP

Applications – Different geometries

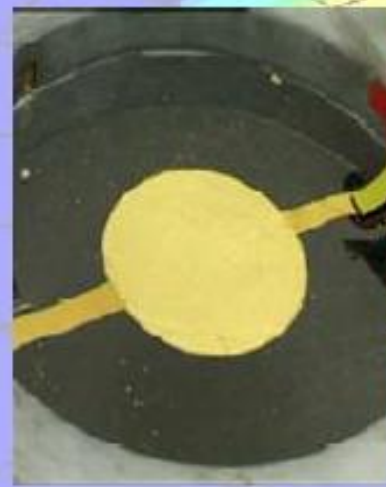
Experiments and Finite Element Models

Several experiments at different prestrain levels have been performed. A constant voltage between 2 and 3.5 kV has been applied.

The constitutive mechanical model is a quasilinear visco-hyperelastic model. This model is implemented in a finite element model (see figure 2) for calculation of the behavior of the circular actuator.

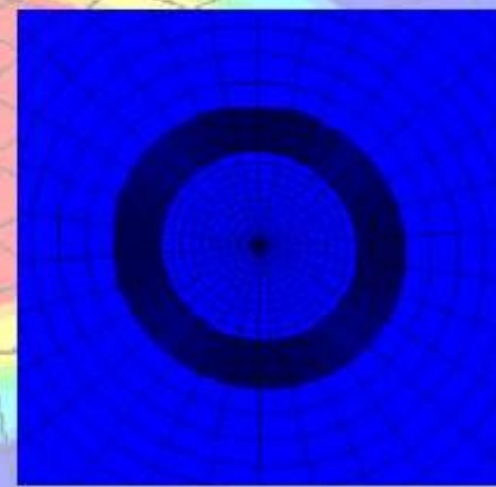


Voltage on

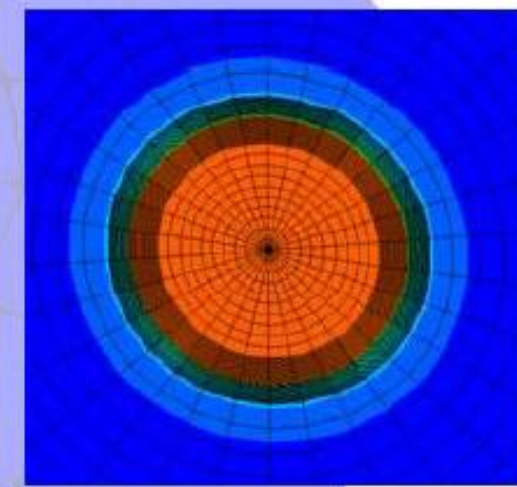


Voltage off

Figure 1: Experimental setup of the circular actuator.



Voltage on



Voltage off

Figure 2: Finite element model of the circular actuator.



Materials Science & Technology



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Dielectric EAP

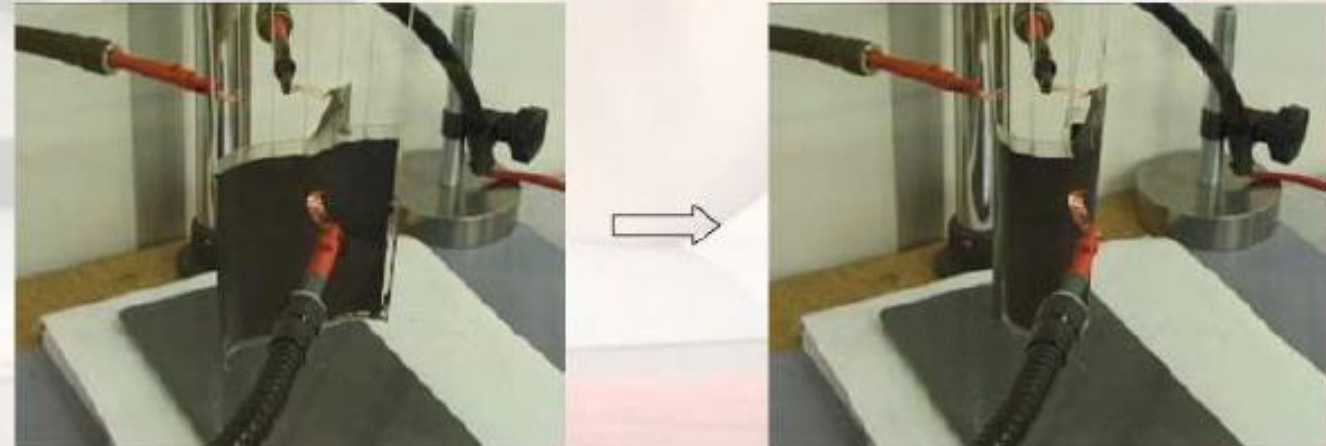
Applications – Different geometries

Configurations and Characteristics of Dielectric Electroactive Polymers

Possible Actuator Configurations

Various actuator designs are possible such as:

- Single-layer or multilayer planar actuators
- Spring roll actuators (electroactive polymer wrapped around a coil spring)
- Shell-like actuators (bending actuators)



Bending actuator in deactivated (left) and activated state (right).

Characteristics of Empa spring roll actuators

- Free strain: 35 %
- Blocked force: 7 N
- Specific work output: 4.3 mJ/g
- Activation cycles: 500



Spring roll actuator.



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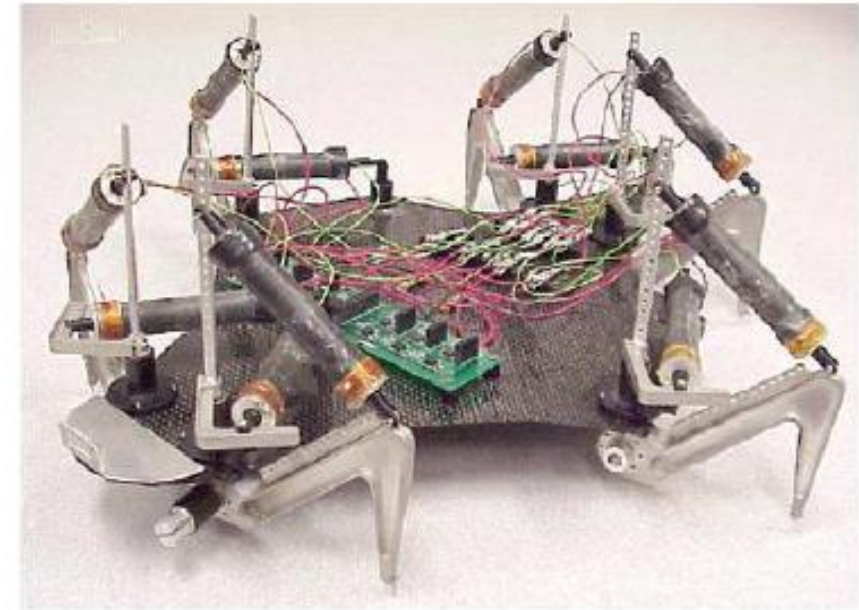
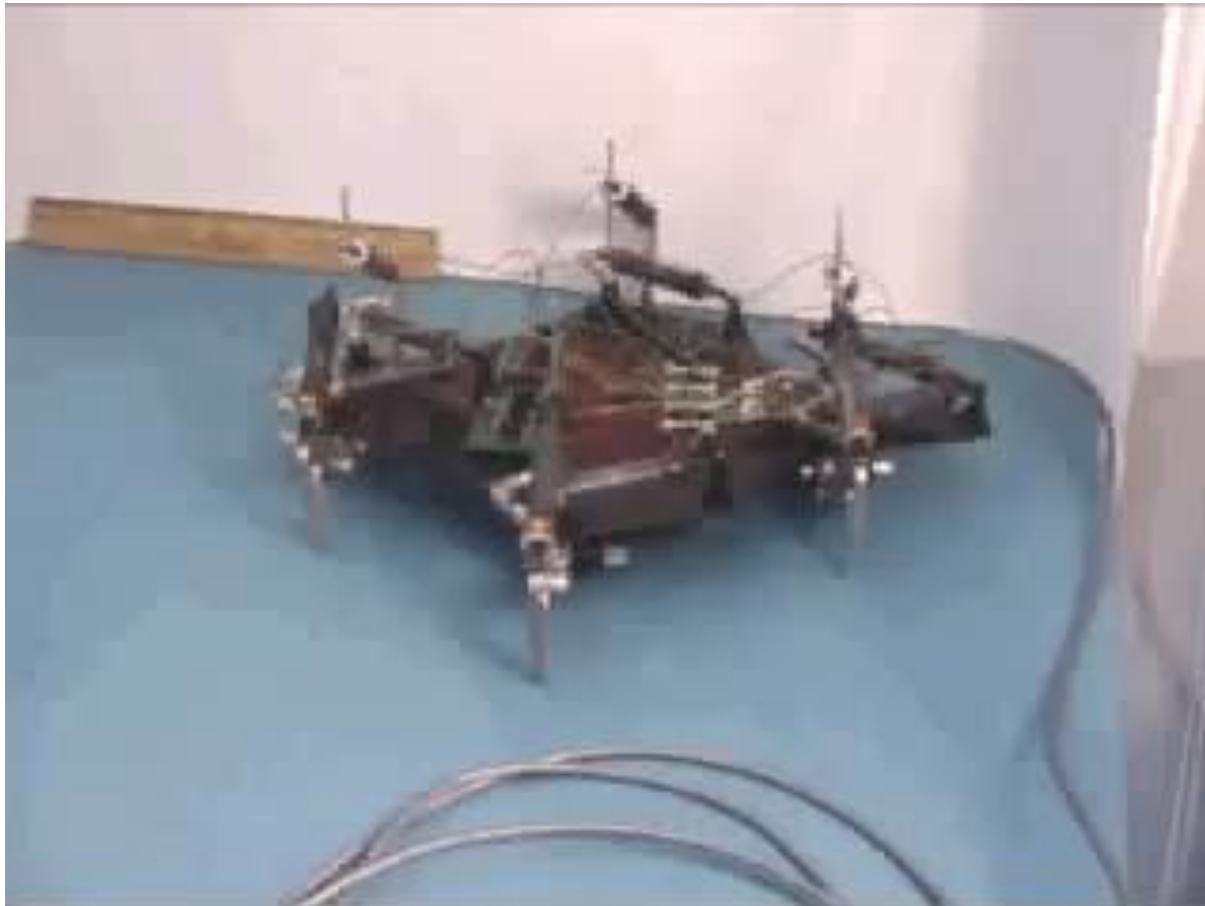
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Swiss Federal Institute of Technology Zurich



Dielectric EAP

Applications – Robotics examples

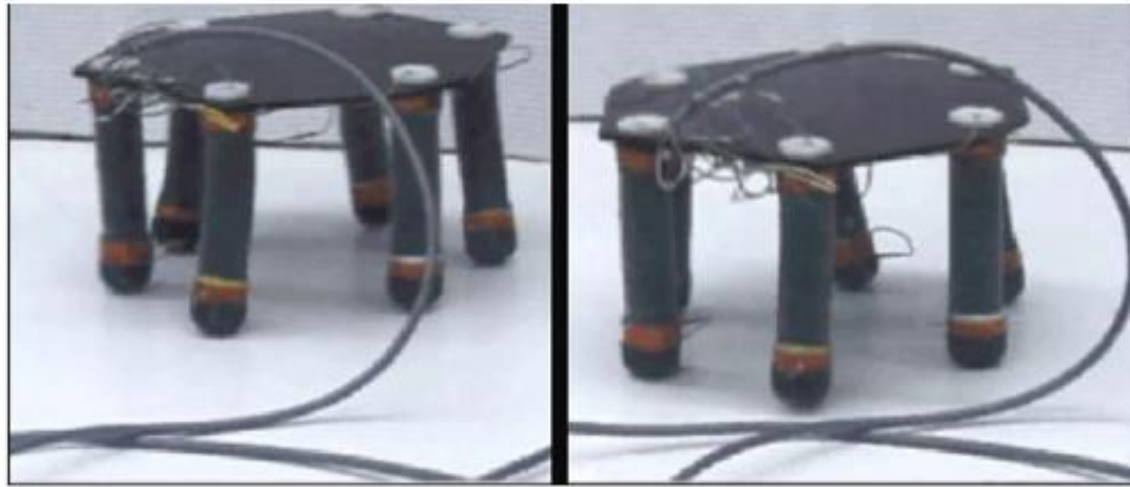
FLEX2 (SRI, International USA)



Dielectric EAP

Applications – Robotics examples

MERbot (SRI, International USA)



[VIDEO MERbot](#)



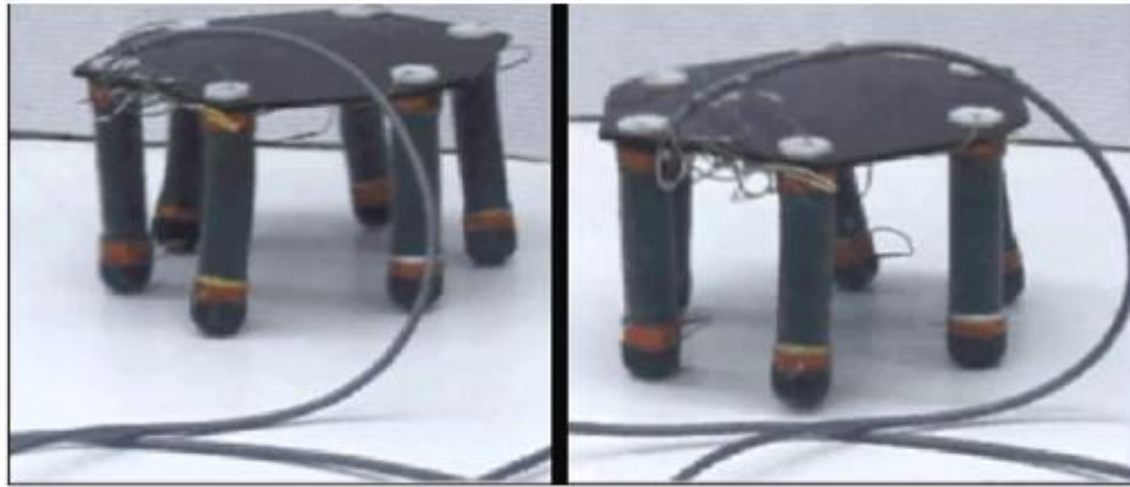
MERbot (Multifunctional ElectroelastomerRolls robot) in action. MERbot is a hexapod robot developed at SRI International to show how dielectric elastomer actuators can greatly simplify robot design. Bending roll actuators provided flexibility, range of motion and simplicity not achievable with any other existing actuation technology. The MERbot was a demonstration of the advantages of replacing legs made out of many parts (links, joints, and actuators) with a single, flexible bending actuator that is both the structure and actuator.



Dielectric EAP

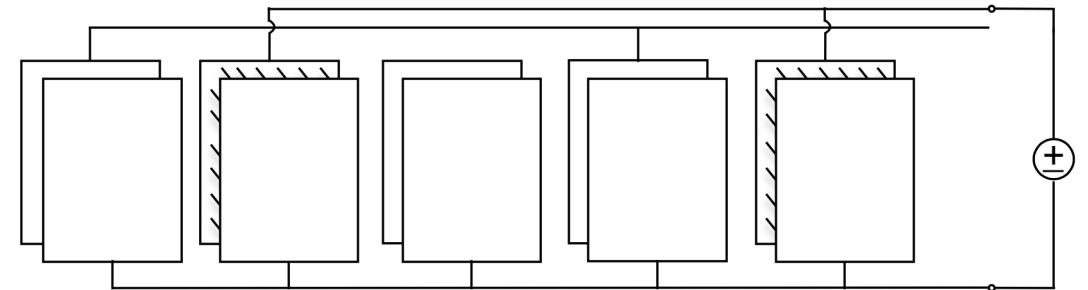
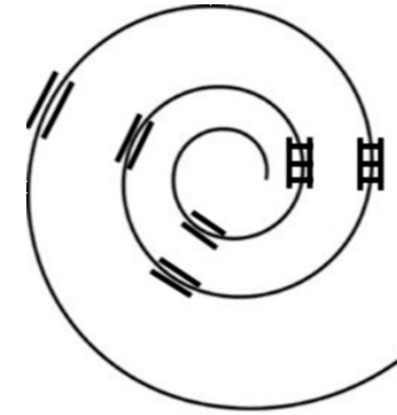
Applications – Robotics examples

MERbot (SRI, International USA)



[VIDEO MERbot](#)

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Dielectric EAP

Applications – Robotics examples

MERbot (SRI, International USA)



[VIDEO MERbot](#)

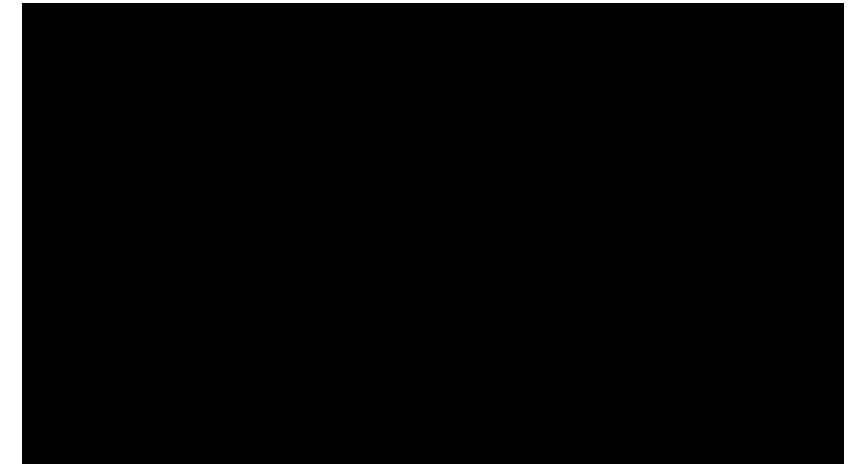
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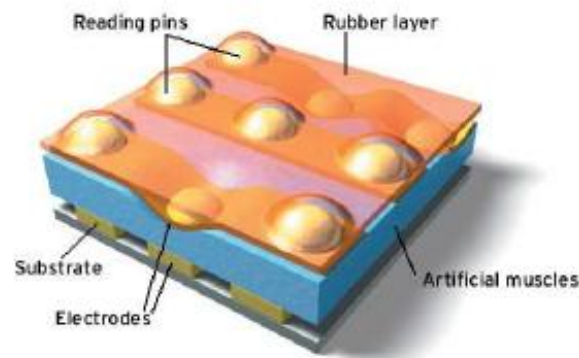
Dielectric EAP

Applications – Other examples

Dirigibile (EMPA)

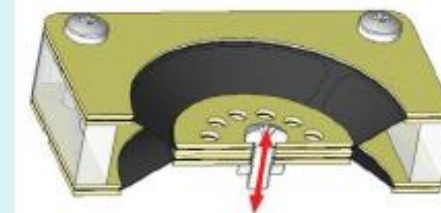
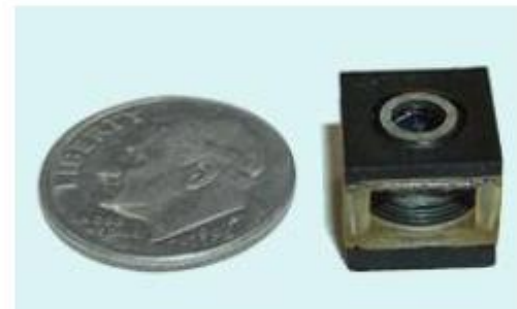


BUBBLE EAP (EPFL)



OTTICA

Autofocus lens positioner
(Artificial Muscles DLP-95)



Dielectric EAP

Applications – Other examples

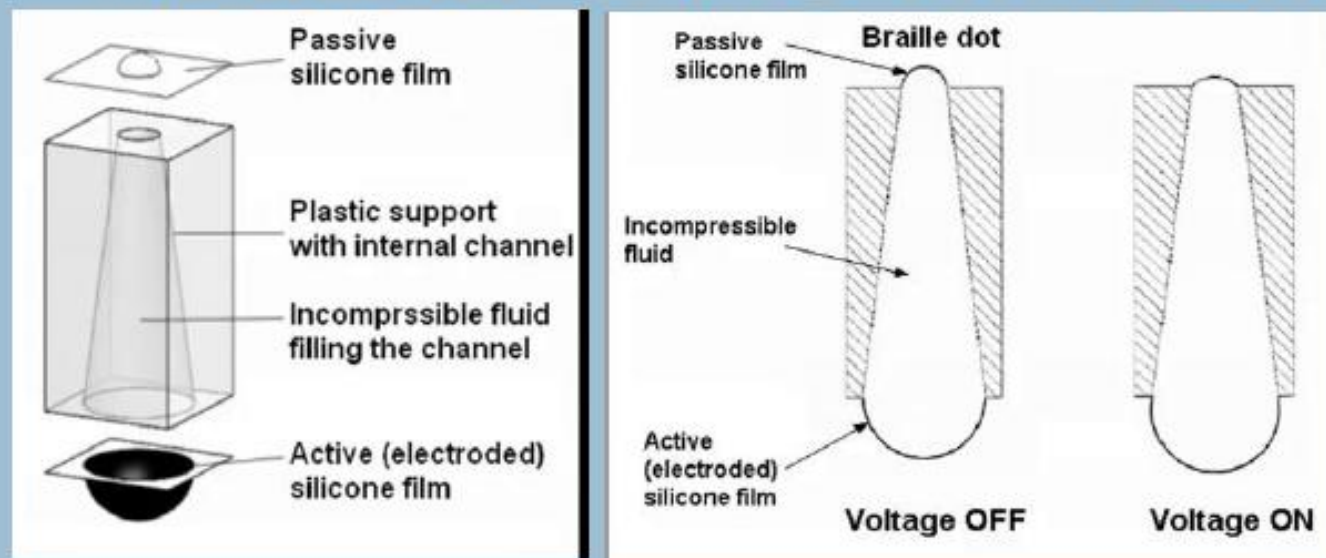
Display Braille basati su DE-EAP



Sungkyunkwan University, South Korea.

Centro Piaggio, Università di Pisa

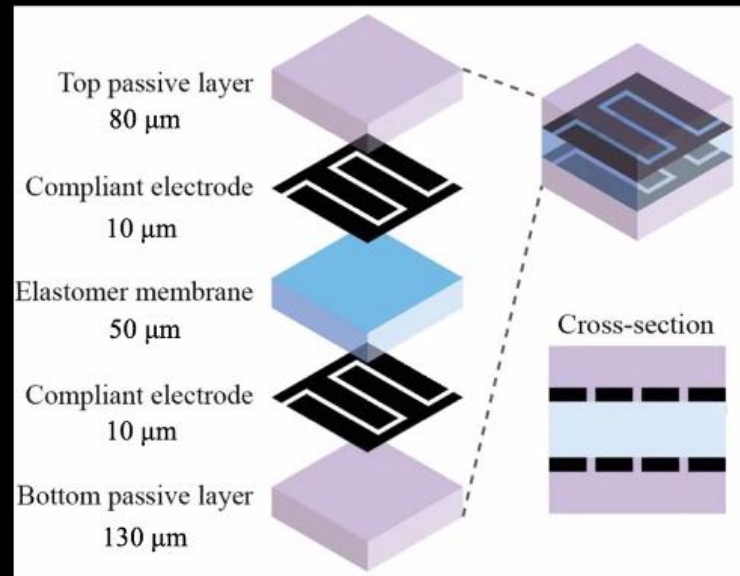
Hydrostatically coupled dielectric elastomer actuator



Dielectric EAP

Applications – Electroadhesive DEA

DEA for bending + Electroadhesion for holding

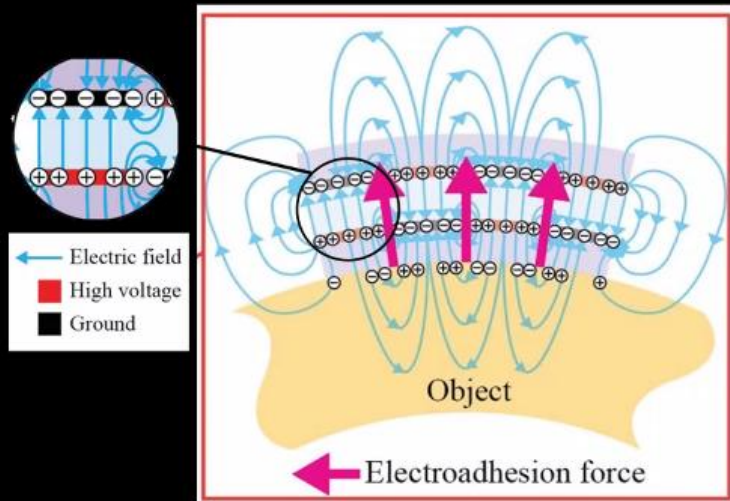


Electroadhesion = high shear force

$$\sigma_{EA} \sim E_{BL}^2 \epsilon_{BL}^2$$



Electroadhesion and DEA



Shintake J, Rosset S, Schubert B, Floreano D, Shea H. Versatile Soft Grippers with Intrinsic Electroadhesion Based on Multifunctional Polymer Actuators. Advanced Materials. 2016;28(2):231–238



Dielectric EAP

PROs and CONs

Benefits	Drawbacks
Large strain (10% - 380%)	High voltages (> 1 KV) and fields (~ 150 MV/m)
Moderate stress (several MPa peak)	Typically required DC-DC converters
Large work density (10k to 3.4 MJ/m^3)	Compliant ($E \sim 1 \text{ MPa}$)
Moderate to high bandwidth (10 Hz to $> 1 \text{ KHz}$)	Pre-stretching mechanisms currently add substantial mass and volume, reducing actual work density and stress
Low cost	
Low current	
Good efficiency ($> 15\%$ typ., 90% Max)	



Definition

Ionic (wet) EAP

- Carbon NanoTubes (CNT)
- Ionic Polymers Gel (IPG)
- Ionomeric Polymer-Metal Composite (IPMC)
- Conductive Polymers (CP)

Advantage	Disadvantage
Requires low voltage	Except for Conductive Polymers, Ionic EAPs don't hold strain under DC voltage
Provides predominately bending actuation (longitudinal mechanisms can be constructed)	Slow response (fraction of a second)
Exhibits large bending displacements	Bending EAPs induce a relatively low actuation force
	Except Conductive Polymers & Carbon Nanotubes, it is difficult to produce a consistent material (particularly Ionomeric Polymer-Metal Composites)
	In aqueous systems the material sustains hydrolysis at $> 1.23V$

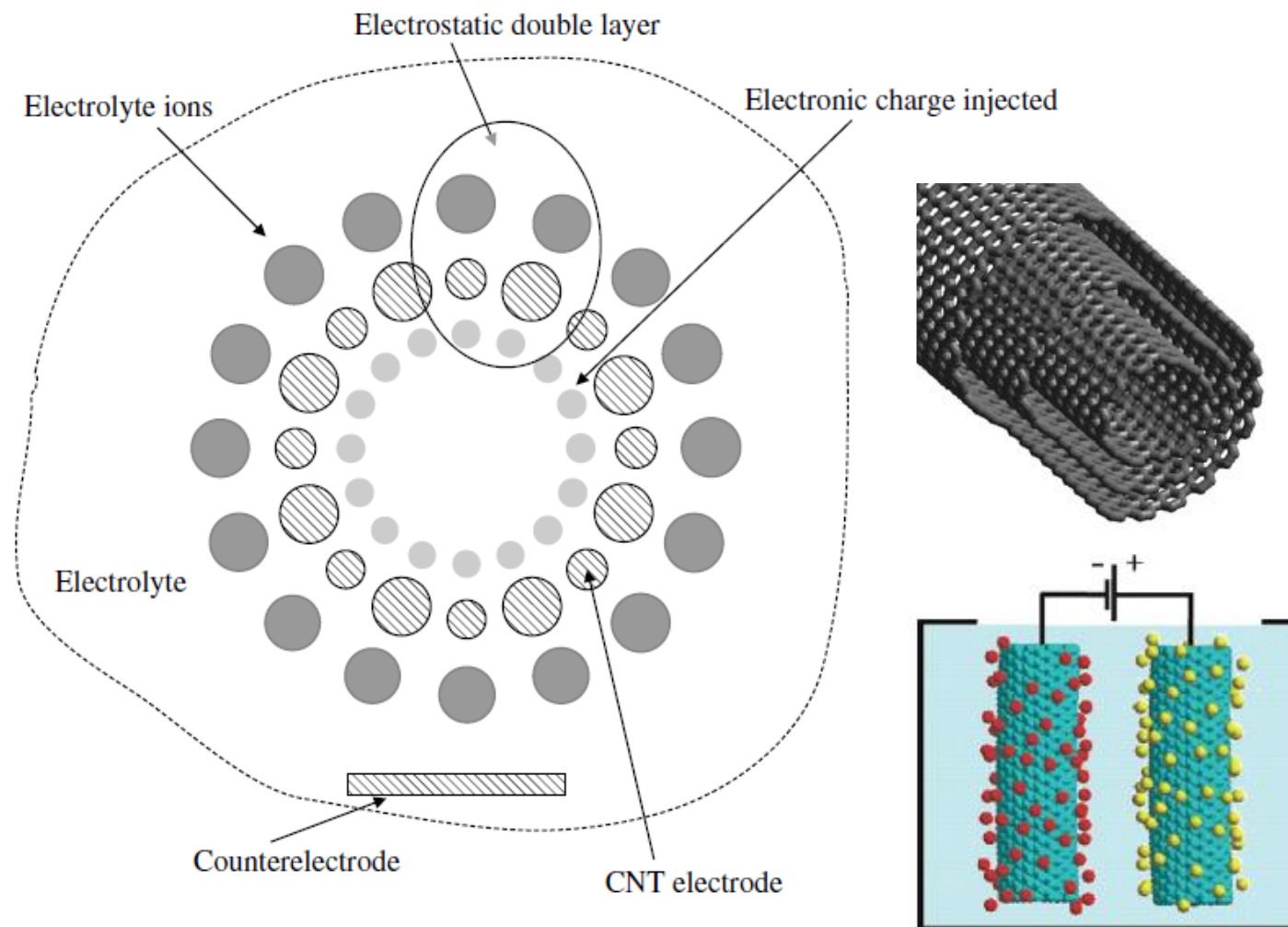


Carbon NanoTubes (CNT)

The actuation principle of operation of CNTs is based on the use of these structures as electrodes in electrochemical supercapacitors.

A voltage applied between an actuating nanotube electrode and a counter electrode (usually another CNT), both dipped in an ion containing solution, leads to charging.

Electrostatic repulsive forces between like charges on the CNTs work against the stiff carbon-carbon nanotubes bonds to elongate and expand the nanotubes.



1. *Quantum mechanical effects.*

These effects are responsible for the expansion of the CNT structure upon injection of electrons and its contraction upon injection of holes.

2. *Quantum chemical effects.*

3. *Electrostatic double layer effects.*

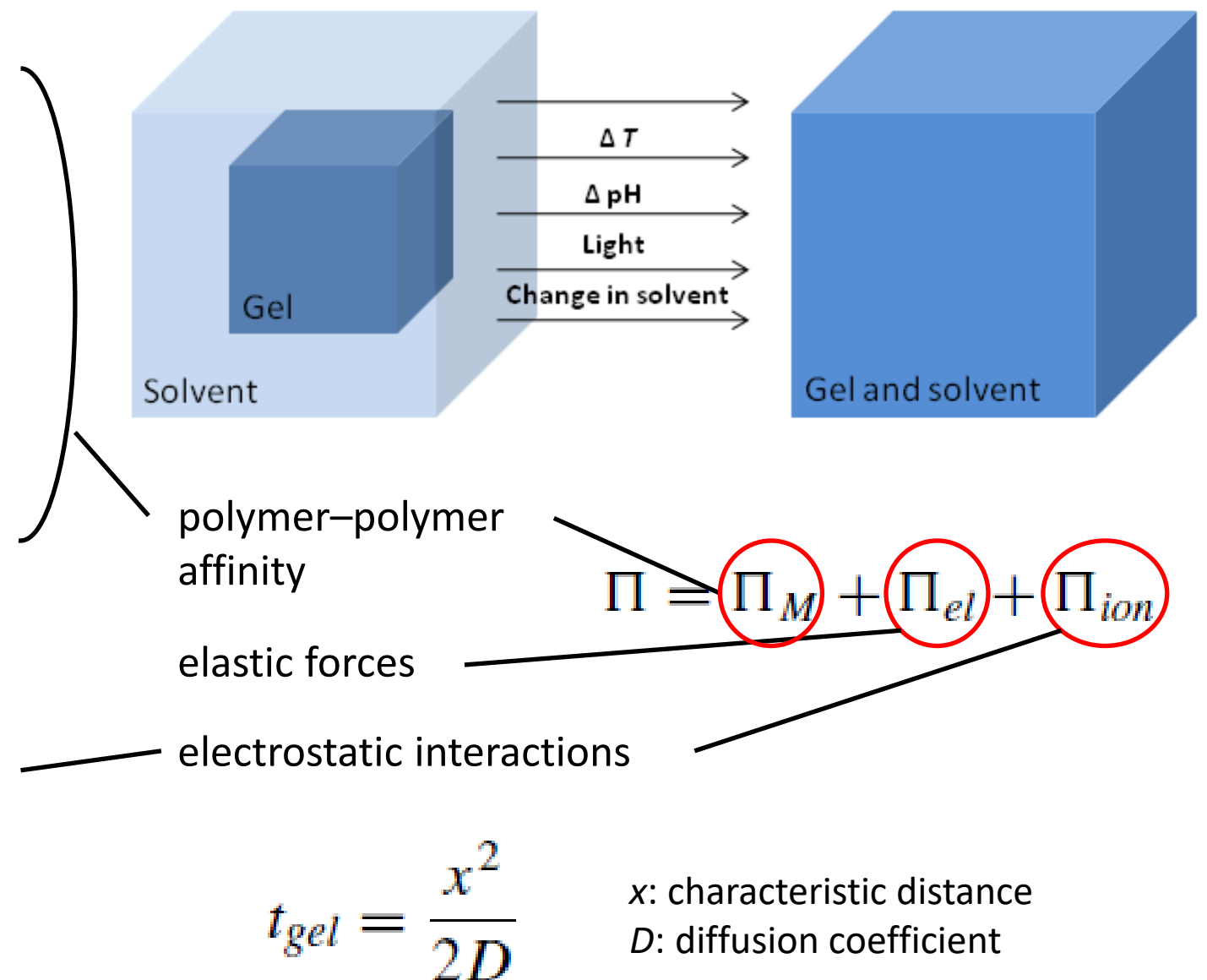
Both quantum chemical effects and double layer effects result in expansion of the CNT structure.



Ionic Polymers Gel (IPG)

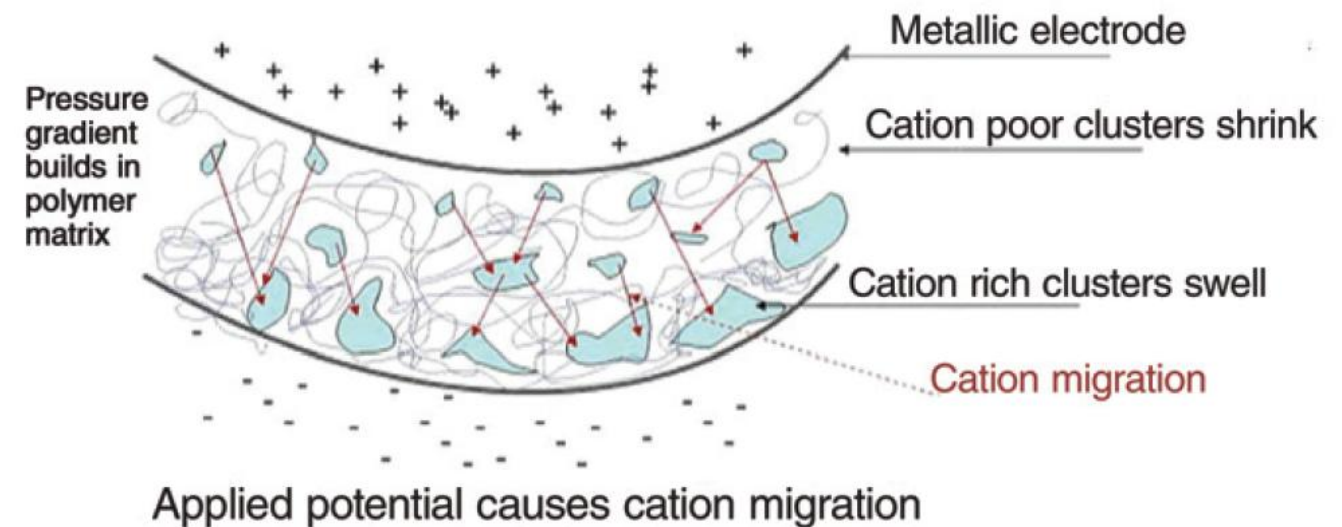
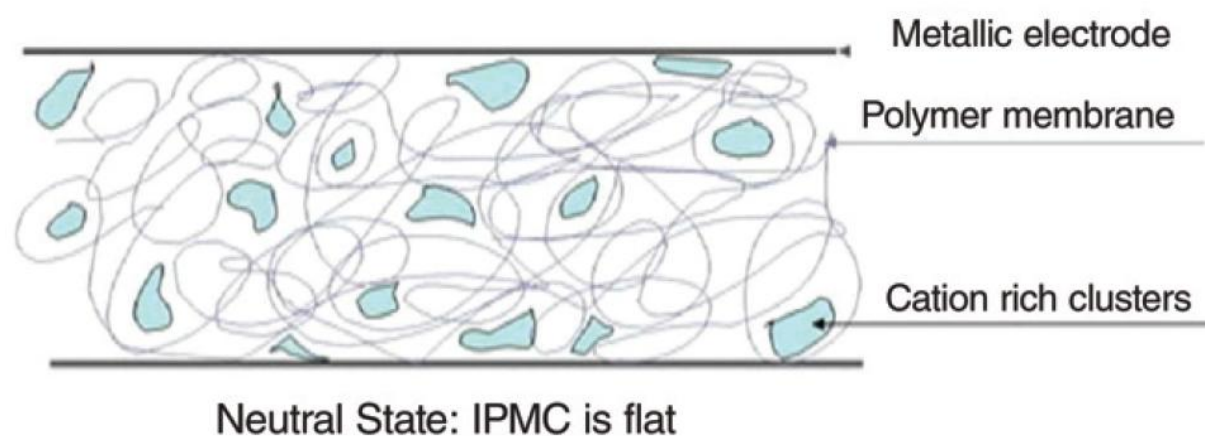
The presence of the solvent inside the polymer network determines the physical properties (stiffness, shape etc) of the gel. By electrical means, polymer gels are able to swell by the uptake of solvent within the polymer matrix.

1. Van der Waals force, resulting from interactions between neighboring molecules in a nonpolar solvent.
2. Hydrophobic interaction, occurring between water molecules in the gel and hydrophobic polymer chains.
3. Hydrogen bonding, resulting from the interaction of hydrogen atoms located between two electronegative atoms.
4. Electrostatic interactions, taking place in the so-called polyelectrolyte, that is, with fixed charges on the network. Ions of opposite charge will be attracted to provide electrical neutrality. The pressure produced by the relative concentration of ions will result in osmotic pressure.

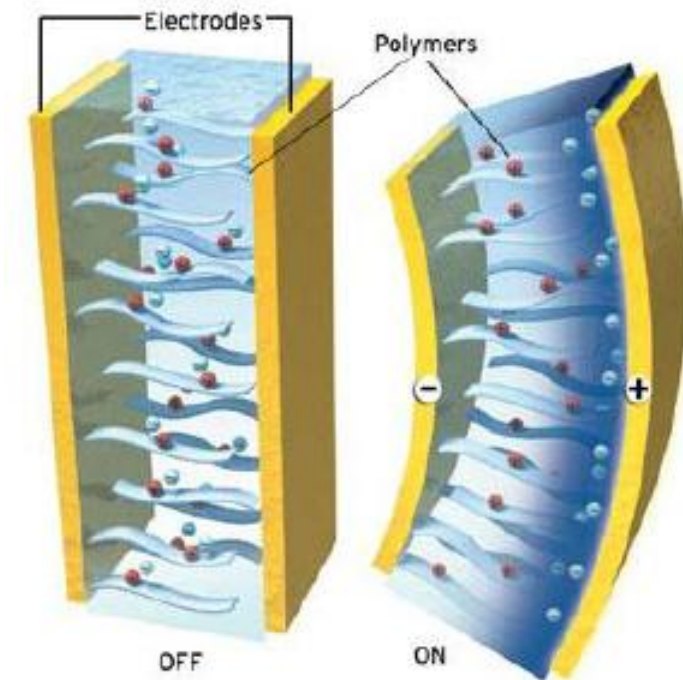
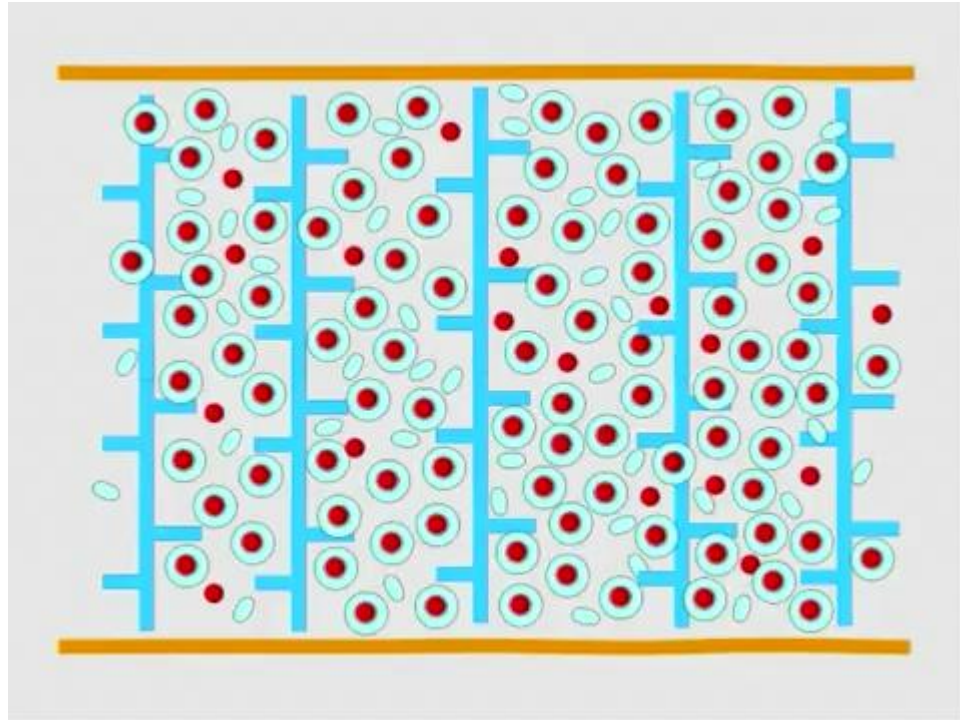


Ionomeric Polymer-Metal Composite (IPMC)

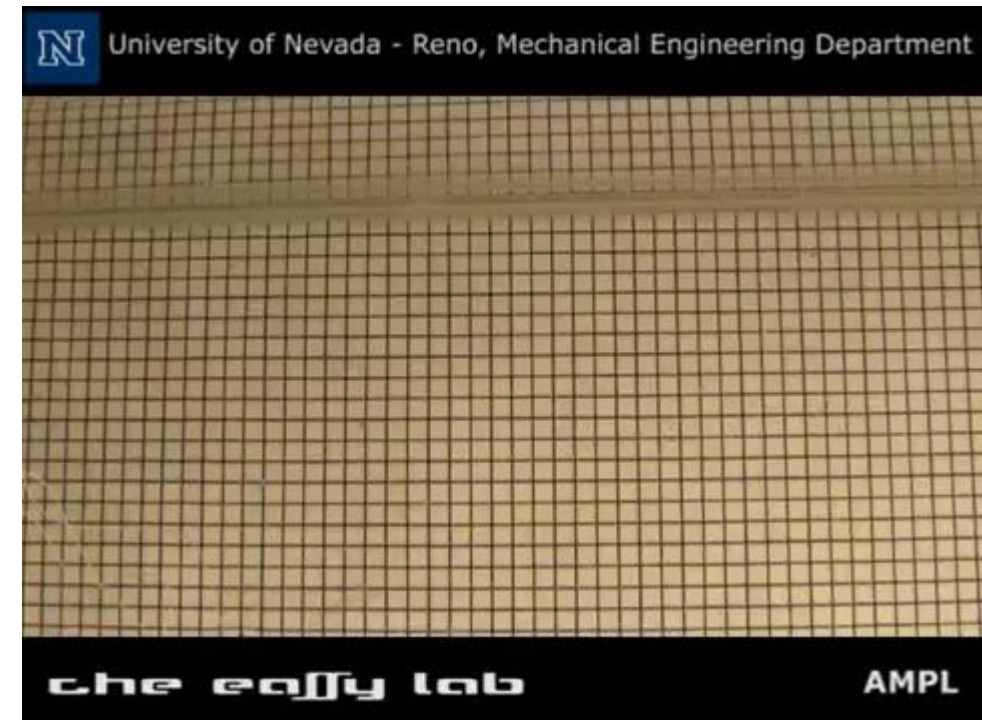
This type of electrochemical artificial muscle employs a polyelectrolyte as an ion exchange membrane (Nafion, Flemion, Aciplex) and metal thin layers as electrodes (platinum, gold). The polyelectrolyte is sandwiched between the two flexible electrodes, which interpenetrate the polyelectrolyte. When a voltage is applied between the electrodes, the counterions tend to diffuse to the corresponding electrode, thus solvated mobile cations move toward the oppositely charged electrode, resulting in a migration that causes the deflection of the multi layered structure toward one of the metal electrode.



Ionomeric Polymer-Metal Composite (IPMC)



Ionomeric Polymer-Metal Composite (IPMC)

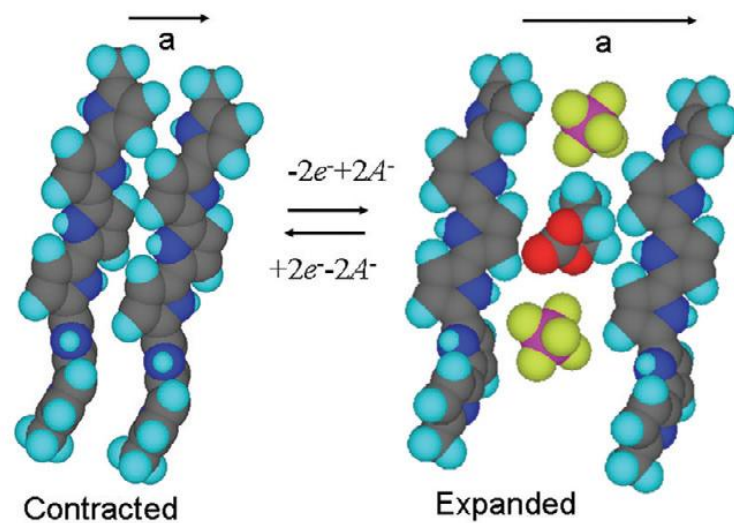


Benefits	Drawbacks
Low voltage (<10 V)	Low efficiency
Large displacement	Low electromechanical coupling
	Requires encapsulation

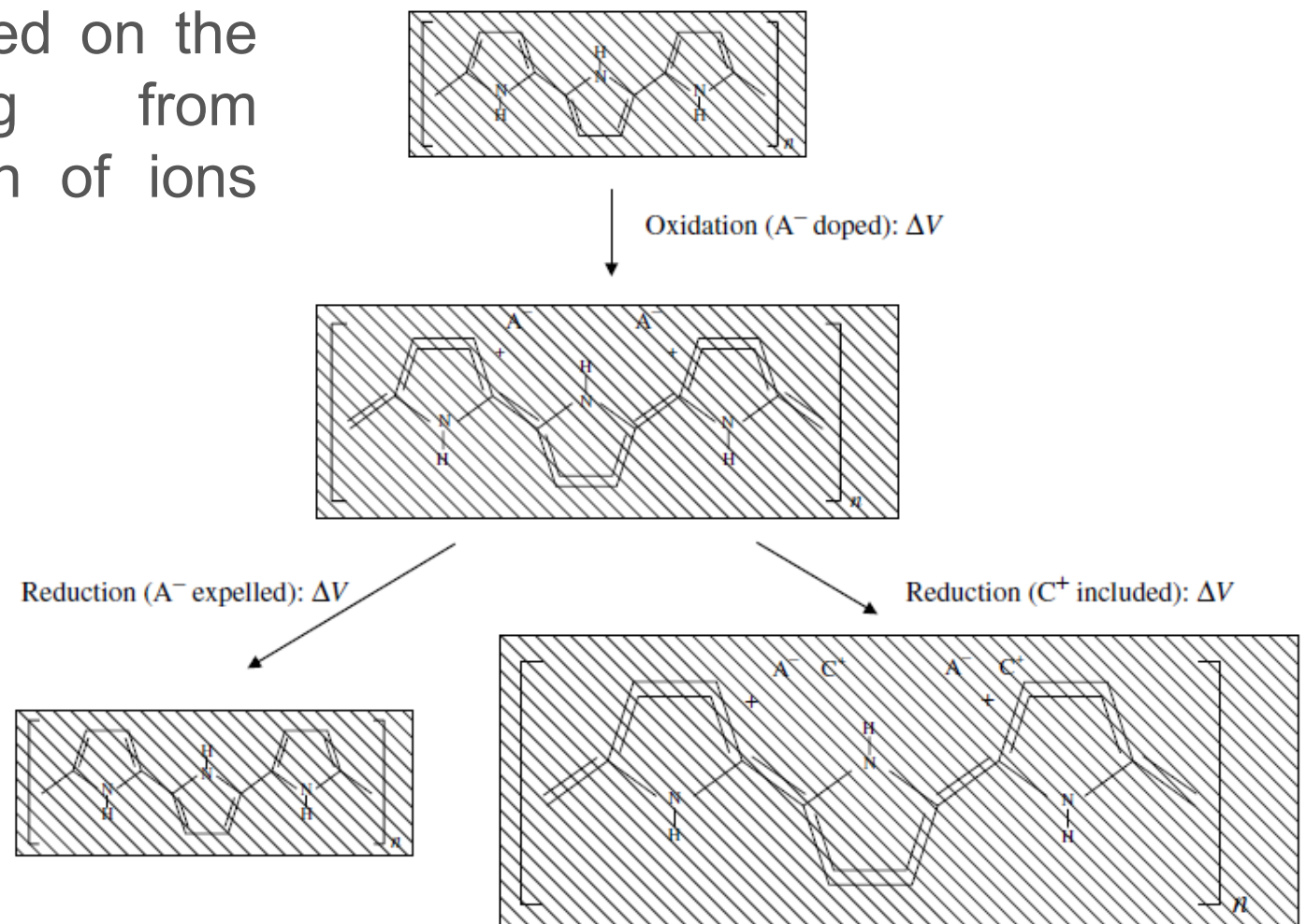


Conductive Polymers (CP)

The mechanism of actuation is based on the dimensional changes resulting from electrochemical uptake or expulsion of ions and solvent



Polypyrrole, Polyaniline, Polythiophene



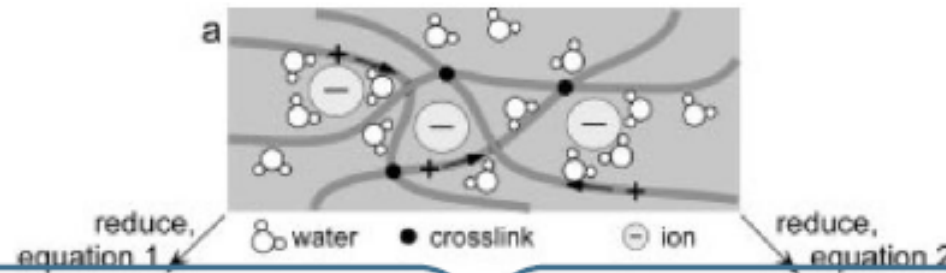
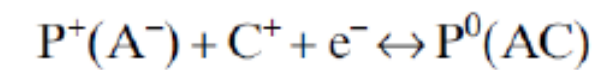
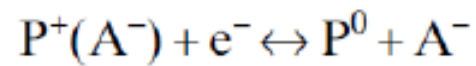
Two types of actuation:

1. The conducting polymer is grown **in the presence of bulky anions**. The elements being taken up and expelled to compensate the electrical charge during the redox process are **cations and solvent molecules**. The volume increases during the reduction process and decreases during oxidation.
2. The conducting polymer is generated **in the presence of small anions**. In this case, compensation of the electrical charge during the redox process is achieved through **expulsion and uptake of anions and solvent molecules**. The volume increases during the oxidation process and decreases during reduction.

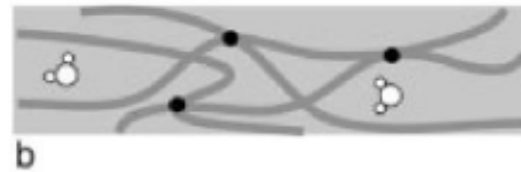


Conductive Polymers (CP)

Oxidized, doped (A-) state



Reduced, uncharged state
(small and mobile anions)

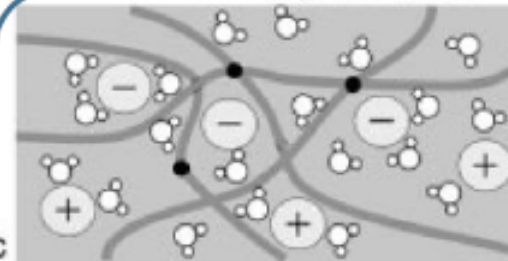


Anions and solvent
(water) are expelled



VOLUME REDUCTION

Reduced state
(large, immobile anions)



Cations and solvent
(water) are incorporated

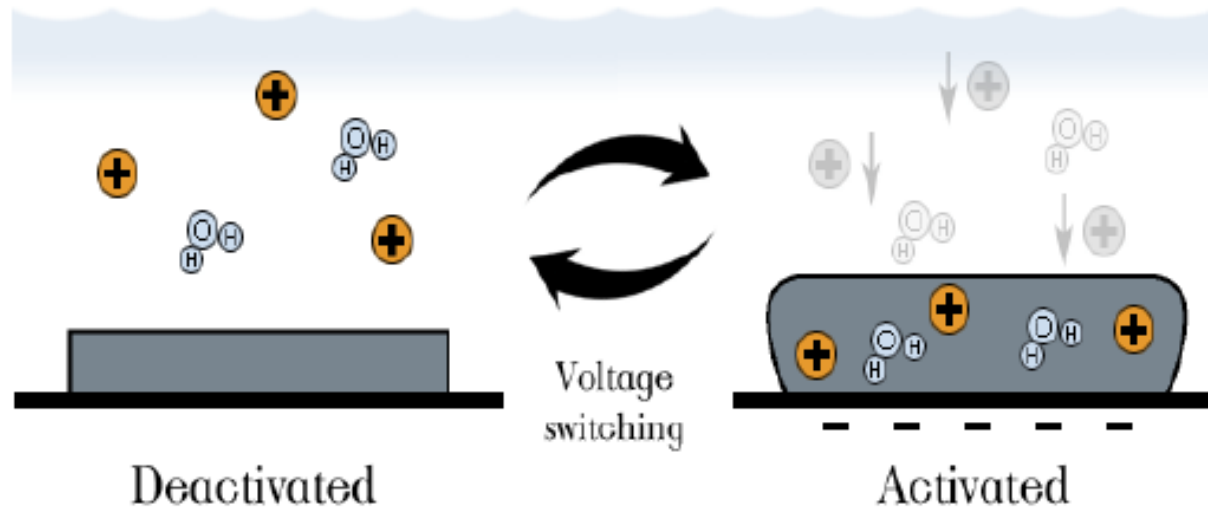


VOLUME EXPANSION



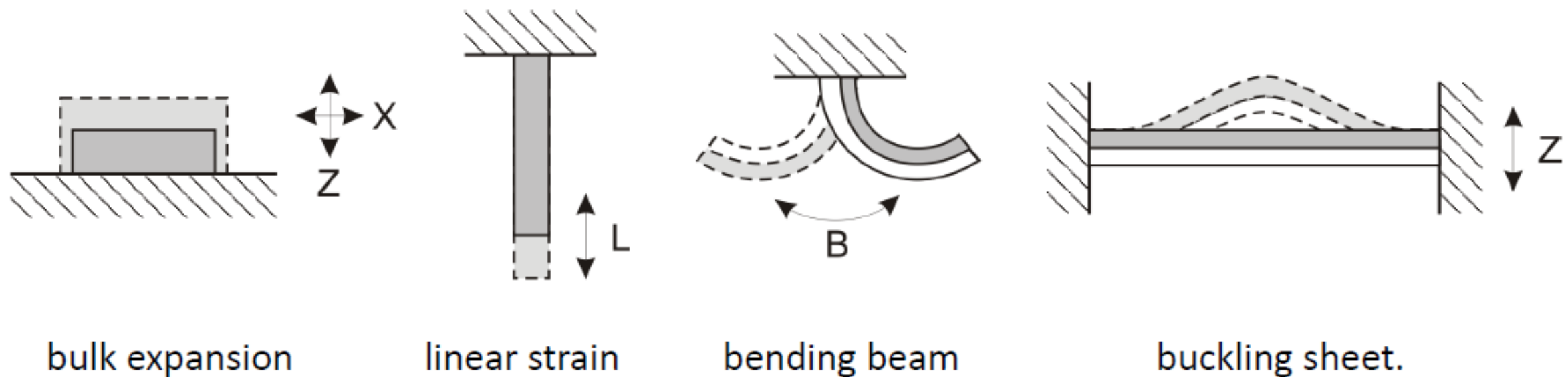
Conductive Polymers (CP)

Technology principle



- Polypyrrole (PPy) swells as a low potential (1-2V) is applied (at low currents)
- Electrolytes: salt solutions, urine, blood plasma, cell culture medium
- Linear 2-5%, perpendicular 30%

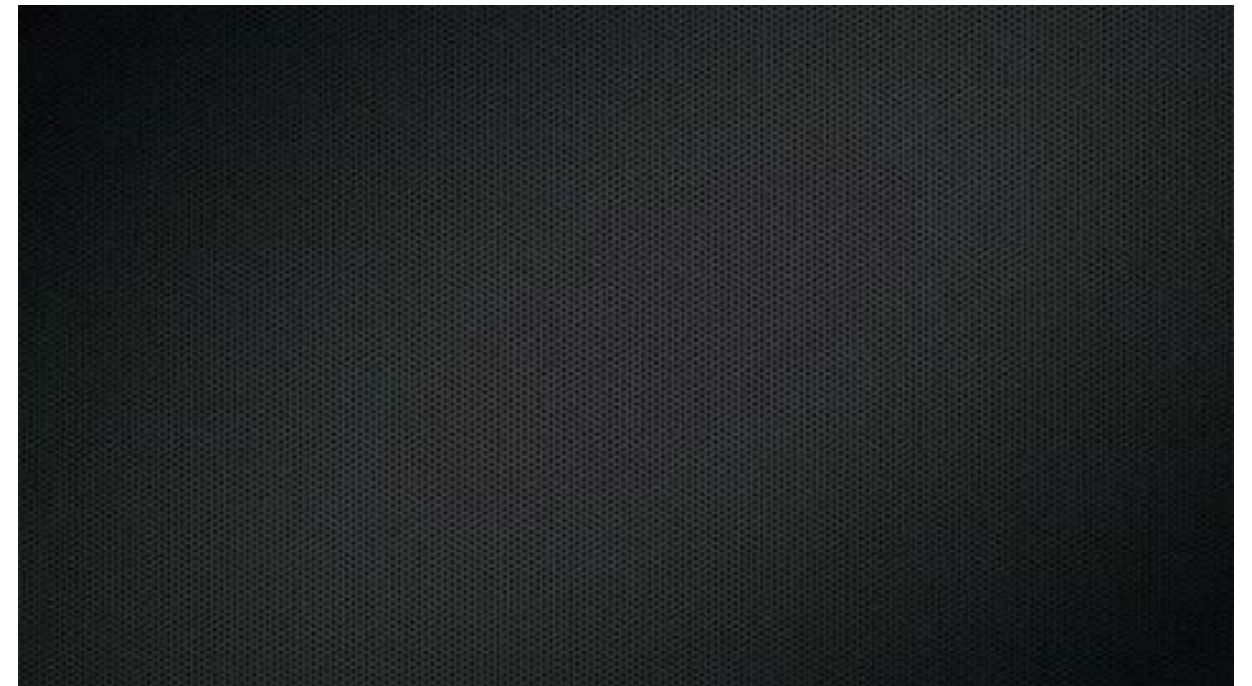
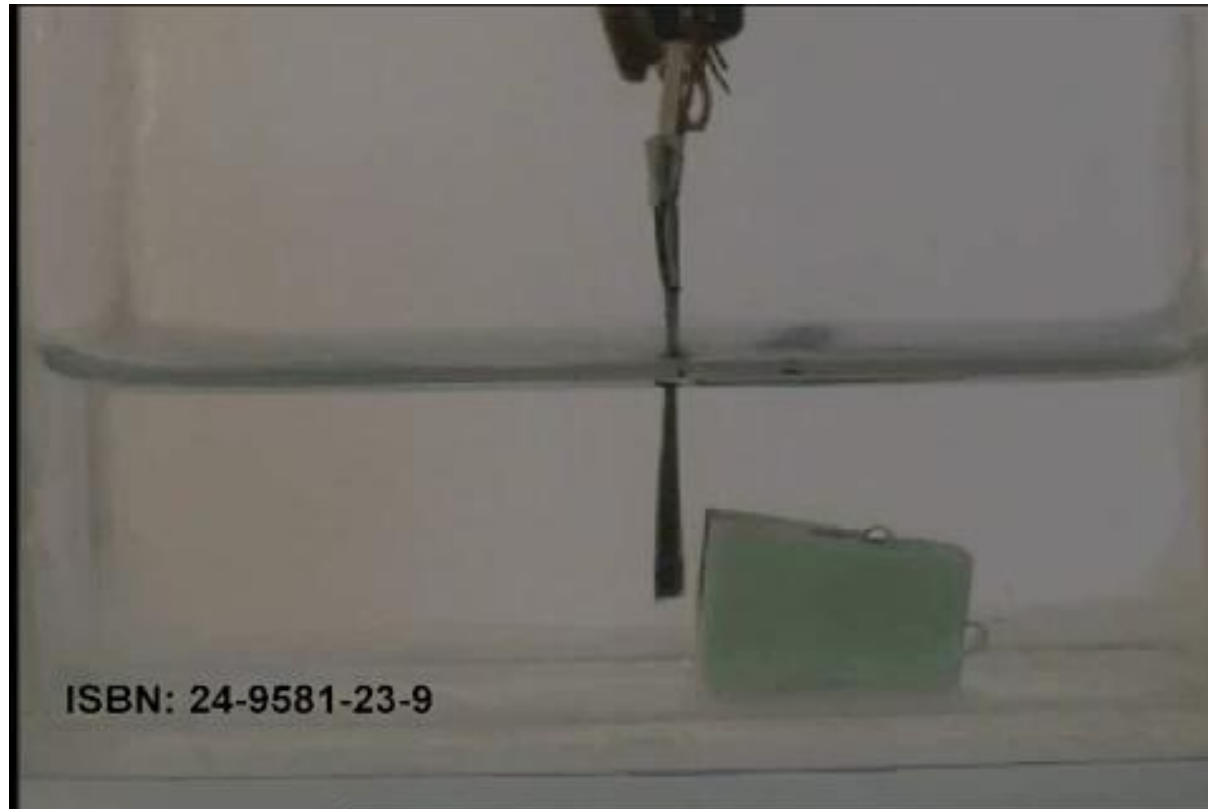
Actuation modes



Courtesy of: E. Jager, Linköping University



Conductive Polymers (CP)



Conductive Polymers (CP)

Benefits	Drawbacks
Moderate to large strains (~3% in plane; >30% out of plane)	Typically need encapsulation
High stress (34MPa max; 5MPa typ.)	Low electromechanical coupling
Large work density (100 kJ/m ³)	Slow (several Hz maximum to obtain full strain)
Stiff polymers (~1GPa)	
Low voltage (1-2V)	



How to choose

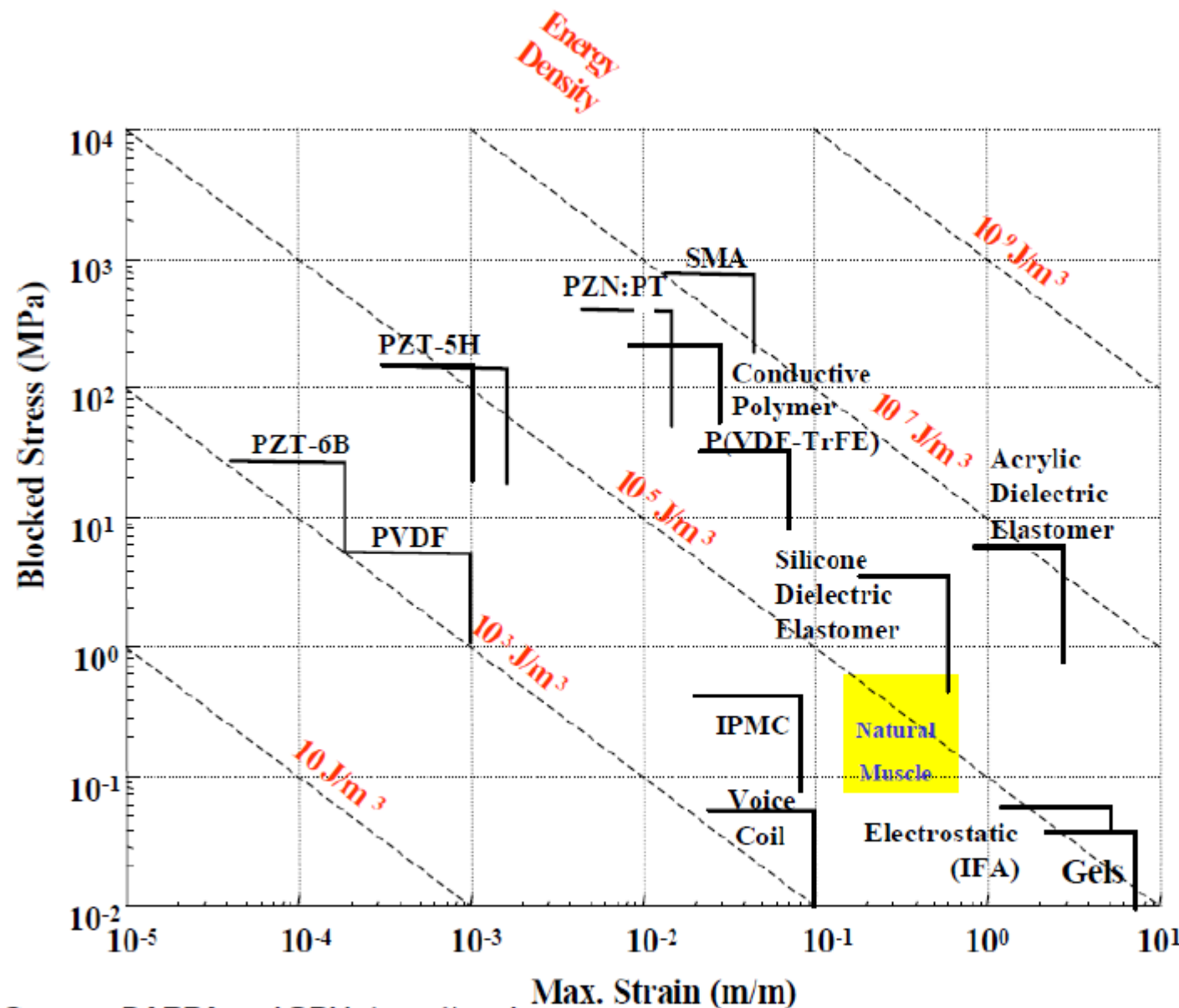
Actuator type	Strain (max) (%)	Stress (max) (MPa)	Work density (max) (J/cm ³)	Specific power (max) (W/g)	Max strain rate (%/s)	Elastic module (MPa)	Max efficiency (%)
Skeletal Muscle	20 (40)	0.1 (0.35)	0.008	0.05	500	10 - 60	40
Dielectric Elastomers	25 (>300)	1.0 (7.0)	0.01 (3.4)	0.5	>450	0.1 – 10	80
Ferroelectric Polymer	3.5 (10)	20 (45)	0.32 (1)	NA	>2000	800	40
Electrostrictive Graft Elastomers	2.5	NA	0.5	NA	NA	600	NA
Liquid Crystal Elastomers	2 (4)	0.12 (0.45)	0.02	NA	1000	100	75
Carbon Nanotubes	<2	>10	1	0.01 (0.27)	19	1000 (sheet) 10000 (fibre)	<1
Conductive Polymers	2 (20)	5 (34)	0.1 (1.0)	(0.1)	10	800 – 3000	1
Ionic Polymer - Metal Composites	0.5 (3.3)	3 (30)	(0.006)	0.003	3.3	50 – 100	1.5 – 3
Active Gels	30	0.2 (1.2)	NA	0.001	NA	<10	NA

<http://eap.jpl.nasa.gov/>



Actuator technology comparison

Energy



Source: DARPA and SRI International

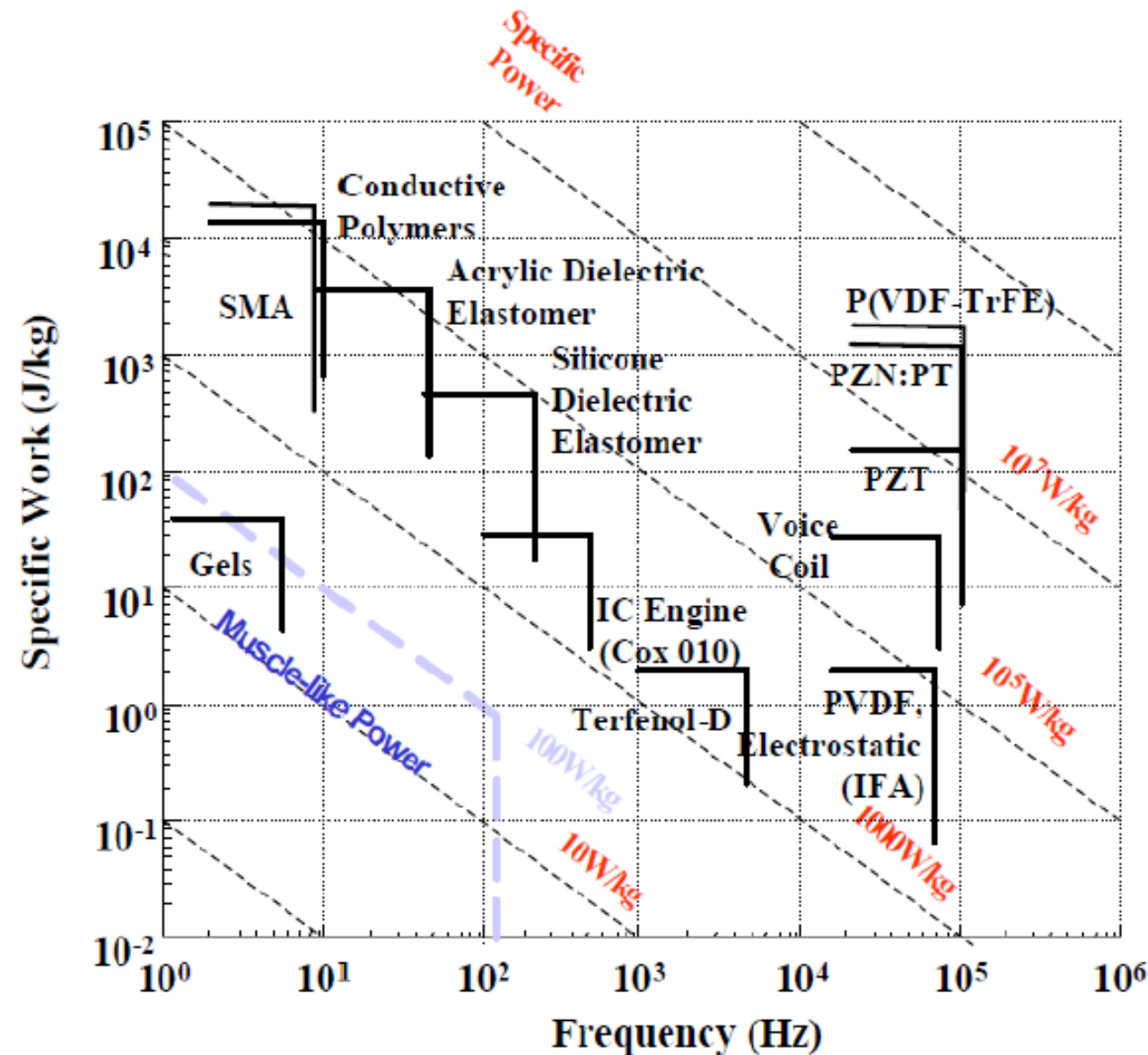
Tables from SRI International (USA), available at <http://ndea.jpl.nasa.gov/nasa-nde/lommas/eap/EAP-web.htm>

- Some polymer actuator technologies have muscle-like performance
- Much of this data is preliminary and for active material mass only
 - practical values can be as much as 10 or 100 times lower



Actuator technology comparison

Power



Source: DARPA and SRI International

Tables from SRI International (USA), available at <http://ndea.jpl.nasa.gov/nasa-nde/lommas/eap/EAP-web.htm>

- Much of this data is preliminary and for active material mass only
 - practical values can be as much as 10 or 100 times lower
- Actuator selection must also consider other factors such as efficiency
- Several smart material or muscle-like actuator technologies look promising considering specific power
 - Field activated materials look promising

